

Analysis of Inverter Output Current Ripple and Design of Inverter-Side Output Filter Inductor for Grid-Connected Applications

Bishal Mondal [✉] and Arun Karuppaswamy B [✉], *Member, IEEE*

Abstract—Incisive selection of the *LCL* filter parameters for a grid-connected inverter (GCI) is crucial to meet the grid interconnection standards with a reduced hardware footprint. Various design methods are available in the literature for selecting the *LCL* filter parameters. While the grid-side inductor of the *LCL* filter can utilize an iron core and follow the standard grid frequency inductor design, the inverter-side inductor design needs attention since it has significant switching frequency harmonics. This paper presents an extensive discussion on the design of the inverter-side inductor for GCIs. The inverter-side inductor (L_i) is calculated based on the allowable inverter peak-peak ripple current to reduce the losses due to the ripple component. The value or size of L_i depends on the inverter configuration, switching technique, and the application. The initial sections of the paper present a comprehensive analysis, comparing the value and hence the size of L_i for different wiring configurations and applications. Closed-form expressions are developed for L_i and are used in selecting the minimum value of L_i . The suitability of an amorphous core for the inverter-side inductor is discussed. The amorphous-core inductor designs in literature can lead to a wide variation of inductance with current and have been analyzed to cause differential and common mode noise. To address this, a novel amorphous-core inductor design is proposed in the later sections of this work. The proposed approach ensures a minimal variation in the inductance over the operating current range. Experimental results are provided to support the various theoretical assertions.

Index Terms—*LCL* filter, AFEC, amorphous core, GCI as DGAs, inverter-side inductor, STATCOM.

NOMENCLATURE

GCI Grid-connected inverter.
 L_i Inverter-side inductor.

Received 13 February 2024; revised 21 May 2024 and 27 August 2024; accepted 2 October 2024. Date of publication 16 October 2024; date of current version 31 January 2025. Paper 2023-SECSC-1877.R2, presented at the 2022 IEEE International Conference on Power Electronics, Drives and Energy Systems, Jaipur, India, Dec. 14–17, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Renewable and Sustainable Energy Conversion Systems Committee of the IEEE Industry Applications Society [DOI: 10.1109/PEDES56012.2022.10080487]. This work supported by New Faculty Scheme, IIT Madras, under Project ELE1617686NFSCAKAR. (Corresponding author: Arun Karuppaswamy B.)

The authors are with the Department of Electrical Engineering, Indian Institute of Technology Madras, Chennai 600036, India (e-mail: ee16d406@ee.iitm.ac.in; akp@ee.iitm.ac.in).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TIA.2024.3481396>.

Digital Object Identifier 10.1109/TIA.2024.3481396

L_g Grid-side inductor.
 R_d Damping resistor.
 C_d Damping branch capacitor.
 C_f Filter capacitor.
AFEC Active front-end converter.
DGAs Distributed generation applications.
SAF Shunt active filter.
VSC Voltage source converter.
PCC Point of common coupling.
 $\Delta I_{i,pk}$ Inverter-side inductor peak-peak ripple.
 $I_{i1,pk}$ Rated fundamental current peak of the inverter.
 f_s Sampling frequency.
 θ_i Phase angle of the inverter-side current.
 v_{L_i} Voltage across L_i .
 v_{PCC} Voltage at the PCC.
 v_i Inverter output voltage.
 f_{sw} Switching frequency.
 V_{dc} DC bus voltage.
 v_{xN} Inverter phase voltage [$x \in (a, b, c)$].
 v_{xO} Inverter pole voltage [$x \in (A, B, C)$].
 v_{NO} Inverter common mode voltage.
 v_{ga} Grid phase voltage (a-phase).
 $v_{C_f,a}$ Voltage across the shunt capacitor C_f (a-phase).
 m_a modulation index (a-phase).
 $v_{g,nom.}$ Nominal grid voltage.
 $B_{sat.}$ Saturation flux density.
 $I_{pk.}$ Peak current through L_i .
 A_e Core cross-section area.
 N Number of turns.
 l_g Air gap length.
 μ_o Permeability of free space.
 μ Permeability of core.
 l_m Magnetic path length.
 A_L Inductance factor.
 $B_{max.}$ Maximum flux density.
 K_u Window utilization factor.
 E Energy stored in inductor.
 W_a Window area.
 $A_{L,pk.}$ Inductance factor at peak current.
 v_x^* Reference voltage [$x \in (a, b, c)$].
 $v_{tri.}$ Carrier signal.
 θ_x^* Reference voltage phase angle [$x \in (a, b, c)$].
TF Transfer function.

I. INTRODUCTION

A WIDE array of GCIs are available today, each serving a unique purpose [1], [2], [3]. A few examples are active front-end converter (AFEC), static synchronous compensator (STATCOM), shunt active filter (SAF), and GCI for distributed generation applications (DGAs). GCIs for power quality improvement and distributed generation applications are basically voltage source converter (VSC) operating in a closed-loop current control mode [4]. The controller generates the required reference voltages for achieving the desired current injection at the point of common coupling (PCC). The reference voltages are compared with a high-frequency carrier signal to generate the switching pulses for the VSC. Therefore, the output voltage of VSC comprises of high-frequency components along with the required fundamental frequency component. The harmonic currents resulting from the inverter's high-frequency voltage components are attenuated using an output filter connected between the GCI and the PCC [5]. The limits on the harmonic current injection are usually specified by grid codes such as the IEEE-1547 standard [6]. The output filter design must ensure that the GCI complies with such standards for grid connection. An LCL filter is capable of meeting the mandated harmonic current requirements with a smaller size [7], [8] and is usually preferred. However, resonance is inherent in LCL filters, and hence damping becomes essential. Passive damping is one of the options. Two major variants of passive damping for the LCL filter are discussed widely in the literature – series resistive (R_d) damping and parallel resistive-capacitive (R_dC_d) damping [9], [10], [11], [12].

The design of a passive damped LCL filter is well discussed in the literature [9], [10], [11], [12], [13]. In a LCL filter based GCI, due to the pulse width modulated nature of the inverter output voltage, the inverter-side inductor current (i_i) comprises of the switching frequency ripple riding on the fundamental component. The peak-peak ripple ($\Delta I_{i,pk.}$) in the inverter-side inductor current usually varies with every switching cycle. The value of L_i in LCL filters, for both R_d and R_dC_d damping, is usually calculated based on the allowable maximum $\Delta I_{i,pk.}$ ($\Delta I_{i,pk.,m}$) [7], [11], [12], [13]. Typically, $\Delta I_{i,pk.,m}$ is selected between 10% to 30% of the inverter's rated fundamental current peak ($I_{i1,pk.}$) [7], [11], [12], [13]. The 10% limit ensures that the fundamental voltage drop across L_i does not exceed 10% of the nominal voltage at the rated current. This is necessary to avoid a high dc bus voltage requirement, which can lead to higher losses in the switches. The 30% limit is set to ensure a lower ripple loss in the switches. Additionally, maintaining a lower ripple ensures that the linear region of the core flux is effectively utilized by the fundamental frequency component of the current.

After finalizing the value of L_i , the next step is to determine the value of the shunt capacitor. The capacitor value is selected such that the fundamental frequency reactive power drawn by it is low, as this reactive power is typically supplied by the inverter. The grid-side inductor (L_g) and the damping resistor (R_d) are selected to meet the required harmonic attenuation. While the component selection of the passive-damped LCL filter is analyzed in the literature, the design of the inductors has not

been discussed in detail. In a well-designed filter that meets the grid code, the grid-side current predominantly consists of only the fundamental frequency component. Consequently, designing L_g is relatively straightforward. It often involves selecting an iron-core inductor following standard design procedures suitable for fundamental frequency operation. However, the design of L_i is more complex due to several factors. A preliminary analysis was done by the authors earlier [14]. In this paper, a more detailed discussion is provided.

Firstly, the design of L_i requires the knowledge of $\Delta I_{i,pk.,m}$. The value of $\Delta I_{i,pk.,m}$ depends on the wiring configuration and on the application of the GCI. In [11], the relationship between L_i and $\Delta I_{i,pk.,m}$ is derived for a 3-phase 3-wire sinusoidal pulse width modulation (SPWM) based AFEC with LCL filter (both R_d and R_dC_d damping) at the zero-crossing ($\Delta I_{i,pk.,0^\circ}$) and at the peak ($\Delta I_{i,pk.,90^\circ}$) of the inverter current. Depending upon the modulation index, either $I_{i,pk.,0^\circ}$ or $\Delta I_{i,pk.,90^\circ}$ is shown to have the maximum peak-peak ripple and is chosen for the computation of L_i accordingly. However, the method discussed in [11] is applicable only for a 3-phase 3-wire AFEC and cannot be applied to a 3-phase 4-wire AFEC. In this work, an attempt is made to analyze the computation of $\Delta I_{i,pk.}$ for a wide range of applications and configurations.

This paper proposes an analytical approach for obtaining the profile of $\Delta I_{i,pk.}$ versus the phase angle of the inverter-side current (θ_{i_i}) for a 3-phase 3-wire and 3-phase 4-wire SPWM based AFEC, GCI for DGAs and STATCOM with an output LCL filter. Once $\Delta I_{i,pk.}$ versus θ_{i_i} is obtained, the expression of $\Delta I_{i,pk.,m}$ is derived from it. The analysis helps in the appropriate selection of L_i as per the desired $\Delta I_{i,pk.,m}$ at the design stage. Further, it can also help in the design of the inverter-side inductor by knowing the maximum value of its instantaneous current, since the flux saturation depends on the instantaneous current. Due to the approximation, $v_{L_i} \approx |v_{PCC} - v_i|$, the selection of L_i based on the allowable inverter ripple current is independent of the other elements of the filter. Therefore, the proposed approach is valid for LCL filters with both R_d and R_dC_d damping.

Secondly, the inductor L_i carries both the fundamental frequency component and the switching frequency harmonics. Therefore, L_i is expected to have a higher bandwidth of operation to act effectively as an inductor at both frequencies (fundamental and switching frequency). The selection of an appropriate type of core material for L_i also becomes important due to the presence of a significant amount of switching frequency harmonics. A balanced choice of core material, considering factors such as the saturation flux density, cost, and core loss, is essential. An amorphous core is suggested for IGBT-based grid-connected inverters and the design procedure for the same is developed in [15] and [16]. However, the permeability of an amorphous core reduces with bias current, which is not considered in [15] and [16] – leading to a high difference in inductance values at 0 A and the peak current. A time-varying inductance when used as a differential mode filter in 3-phase systems, is reported to introduce differential-mode (DM) and common-mode (CM) noise [17]. Therefore, it is desired that the inductance is less sensitive to the current bias. In this work, a

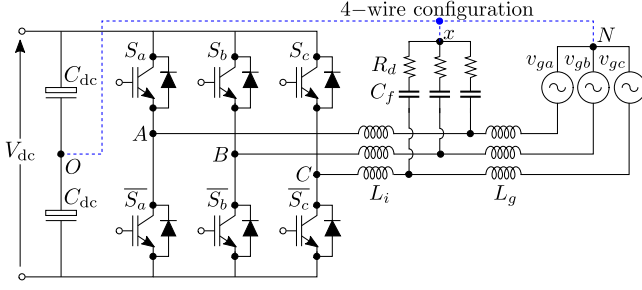


Fig. 1. 3-phase 2-level VSC with an output LCL filter (R_d damping) as a GCI in a 3-wire configuration (solid lines) and 4-wire configuration (dashed lines).

novel amorphous-core inductor design procedure is developed to reduce the variation of inductance with current bias.

The main contributions of this work are listed below:

- 1) *Analytical approach for profile of $\Delta I_{i,pk}$* : An analytical approach is proposed to obtain the profile of $\Delta I_{i,pk}$ versus θ_{i_i} for 3-phase 3-wire and 3-phase 4-wire based AFEC, GCI for DGAs and STATCOM. This approach helps to understand the ripple current pattern for different configurations and applications of GCIs.
- 2) *Closed-form expressions for $\Delta I_{i,pk,m}$* : From the $\Delta I_{i,pk}$ versus θ_{i_i} profile, an expression of $\Delta I_{i,pk,m}$ is derived. The analysis is used for the appropriate selection of L_i to achieve a desired $\Delta I_{i,pk,m}$ at the design stage.
- 3) *Comparison of L_i for 3-wire and 4-wire configurations*: It is demonstrated that a 4-wire configuration requires twice the value of L_i as compared to a 3-wire configuration for the same $\Delta I_{i,pk,m}$, dc-bus voltage (V_{dc}) and switching frequency (f_{sw}). This insight assists in selecting between the GCI configurations concerning the size of L_i .
- 4) *Discussion on core selection*: The choice of core for L_i is discussed in detail outlining the specific requirements of the inverter-side inductor for an output LCL filter in the GCI applications.
- 5) *Modified area product method*: A modified area product method is proposed for the design of an amorphous-core inductor such that it exhibits a minimum variation in inductance with current bias.

The proposed analyses and methodologies are validated through simulation and experimental results.

II. INVERTER-SIDE PEAK-PEAK RIPPLE CURRENT ($\Delta I_{i,pk}$) ANALYSIS OF 3-PHASE AFEC

An AFEC is used to obtain a constant dc voltage at the output while maintaining a unity power factor (UPF) at the ac input side. The input ac currents should also comply with the grid standards. Fig. 1 shows the general structure of a 3-phase 2-level voltage source converter (VSC) connected to the grid through an LCL filter in both 3-wire (solid lines) and 4-wire (dashed lines) configurations. With appropriate controls, the GCI shown in Fig. 1 can be operated as an AFEC [5]. Fig. 2 shows the per-phase representation of an AFEC (valid for GCIs as DGAs and STATCOM) and the corresponding approximated equivalent circuit for understanding the voltage waveform v_{L_i}

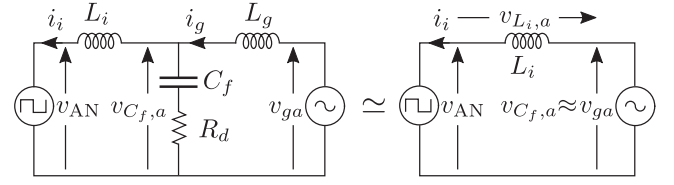


Fig. 2. Per-phase (for phase-a) representation of an AFEC (3-phase 3-wire and 3-phase 4-wire) with an output LCL filter and its approximated equivalent circuit for estimating $v_{L_{i,a}}$.

across L_i . v_{L_i} is required for obtaining the profile of $\Delta I_{i,pk}$ vs. θ_{i_i} . For a 4-wire configuration, obtaining the equivalent circuit as shown in Fig. 2 is simple, since the capacitor voltages are referenced to the grid neutral. On the other hand, for a 3-wire configuration, the capacitor voltages are not referenced to the grid neutral. However, it can be shown that $v_{xO} = v_{NO}$ in a 3-wire configuration, and therefore the equivalent circuit of Fig. 2 is valid for a 3-wire configuration also. It may further be noted that while the equivalent circuits are the same, for a 4-wire configuration, $v_{AN} = v_{AO}$, whereas for a 3-wire configuration $v_{AN} = v_{AO} - v_{NO}$.

For a well-designed LCL filter, the voltage drop across L_g being negligible compared to the PCC voltage [11]

$$v_{C_f,a} \approx v_{ga}. \quad (1)$$

The approximated equivalent circuit in Fig. 2 is obtained based on assumption (1). From the approximated equivalent circuit in Fig. 2, the voltage drop across L_i for a -phase is

$$v_{L_{i,a}} \approx v_{ga} - v_{AN}. \quad (2)$$

The profiles of $\Delta I_{i,pk}$ versus θ_{i_i} are different for 3-wire and 4-wire configurations and are analyzed separately in the following sections. It may be noted that though a simple R -damping is shown in Fig. 1, the method is equally valid for RC -damping and other higher-order damping schemes that use damping elements in the shunt capacitor branch.

A. 4-Wire Configuration

In a 4-wire configuration, the midpoint O of the dc-bus is connected to the neutral point N of the grid (Fig. 1) and $v_{AN} = v_{AO}$. So, v_{L_i} depends only on the inverter pole voltage of the phase considered and the corresponding grid voltage.

The design of the LCL filter usually ensures that the voltage drop across $L_i + L_g$ at the rated current operation is not more than 10% of the nominal grid voltage ($V_{g,nom.}$) [5], [7], [13]. Therefore, the inverter fundamental voltage, which is the phasor sum of the grid nominal fundamental voltage and the voltage drop across $L_i + L_g$, can be assumed to be approximately equal to the grid nominal fundamental voltage itself. With this assumption, the a -, b -, and c -phase reference voltages of the inverter can be written as

$$\begin{aligned} v_a^* &= m_a \sin(\theta_a^*), v_b^* = m_a \sin(\theta_a^* - 120^\circ), \text{ and} \\ v_c^* &= m_a \sin(\theta_a^* + 120^\circ) \end{aligned} \quad (3)$$

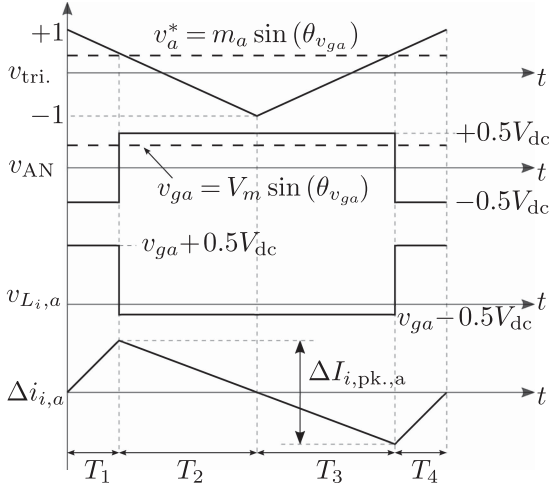


Fig. 3. Plot of v_a^* with v_{tri} , v_{AO} with v_{ga} , $v_{L_{i,a}}$ and $\Delta i_{i,a}$ for one switching period (T_{sw}) for the 3-phase 4-wire AFEC.

where

$$m_a \approx \frac{\sqrt{2}V_{g,nom.}}{0.5V_{dc}} \text{ is the modulation index and } \theta_a^* \approx \theta_{v_{ga}}.$$

Using (3), Fig. 3 shows the plot of a -phase reference voltage v_a^* with carrier signal v_{tri} , inverter pole voltage v_{AO} , a -phase inductor voltage $v_{L_{i,a}}$ and inverter current ripple $\Delta i_{i,a}$ for one switching period $T_{sw} = T_1 + T_2 + T_3 + T_4$. Considering the small duration, the grid voltage v_{ga} is assumed constant during a switching time period.

From Fig. 3

$$\frac{\Delta I_{i,pk.,a}}{2} = \left(\frac{v_{ga} + 0.5V_{dc}}{L_i} \right) T_1 \quad (4)$$

where

$$T_1 = \frac{T_{sw}}{4}(1 - v_a^*)$$

and

$$v_{ga} = \sqrt{2}V_{g,nom.} \sin \theta_{v_{ga}}.$$

In an AFEC, $\theta_{v_{ga}} \approx \theta_{i_{ia}}$. Using $\theta_{v_{ga}} \approx \theta_{i_{ia}}$ and (4), the simplified equation for $\Delta I_{i,pk.}$ for a -phase of the 3-phase 4-wire AFEC is obtained as

$$\Delta I_{i,pk.,a} = \frac{1}{2f_{sw}L_i} \left[\frac{V_{dc}}{2} - \sqrt{2}m_aV_{g,nom} \sin^2(\theta_{i_{ia}}) \right] \quad (5)$$

where $f_{sw} = 1/T_{sw}$.

From (5), it can be observed that the profile of $\Delta I_{i,pk.,a}$ for $\theta_{i_{ia}} = 180^\circ$ to 360° is same as the profile of $\Delta I_{i,pk.,a}$ for $\theta_{i_{ia}} = 0^\circ$ to 180° . Fig. 7 shows the plot of $\Delta I_{i,pk.,a}$ versus $\theta_{i_{ia}}$ for a 3-phase 4-wire AFEC. The calculated values are obtained from (5). It is observed that the analytical and simulation results match well with each other.

B. 3-Wire Configuration

The 3-wire configuration is shown by solid lines in Fig. 1. In this configuration, the inverter voltage referenced to the grid

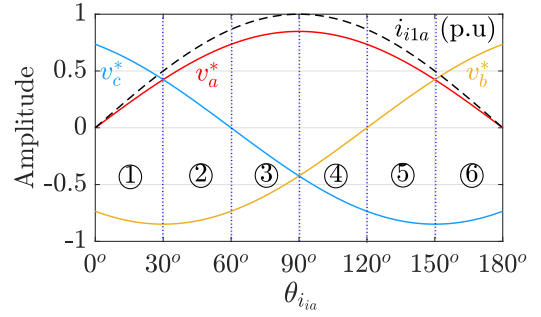


Fig. 4. Inverter reference voltages (v_a^* , v_b^* and v_c^*) and the inverter side fundamental current (i_{i1a}) in p.u. of a 3-phase 3-wire AFEC.

neutral comprises of the inverter pole voltage and the inverter common-mode voltage (v_{NO}). For a -phase, the inverter voltage referenced to the grid neutral in case of a 3-phase 3-wire configuration is

$$v_{AN} = v_{AO} - v_{NO} \quad (6)$$

where

$$v_{NO} = \frac{v_{AO} + v_{BO} + v_{CO}}{3}.$$

From (2) and (6), in a 3-phase 3-wire AFEC, $v_{L_{i,a}}$ for a certain phase is dependent on the corresponding grid voltage and all the three pole voltages of the inverter. A single expression for $\Delta I_{i,pk.}$ is not possible for a 3-wire configuration due to the dependence of v_{AN} on v_{NO} . The pattern of v_{NO} and hence v_{AN} results from the relative values of the three-phase reference voltages. For example, as observed from Fig. 4, in interval (1), $v_c^* > v_a^* > v_b^*$. This leads to the v_{AN} pattern shown in Fig. 5(a) for a sample switching time period in the interval (1). The v_{AN} pattern remains the same for any switching time period in a given interval, although the individual time intervals T_1 , T_2 , T_3 , and T_4 may vary. The v_{AN} pattern will be different in the intervals (2) and (3) since the relative values of v_a^* , v_b^* , and v_c^* are not the same in each interval. The v_{AN} patterns for a sample switching time period for intervals (2) and (3) are shown in Fig. 5(b) and (c) respectively. So, in a 3-wire configuration, computing $\Delta I_{i,pk.}$ requires segregating the inverter current phase angle $\theta_{i_{ia}}$ into intervals of similar reference voltage patterns.

Fig. 4 shows the reference voltages for the 3-phase 3-wire AFEC along with the inverter side fundamental current (assuming the approximate equivalent circuit of Fig. 2). Six different intervals are marked in Fig. 4 based on the relative amplitudes of the reference voltages. The patterns of v_{AN} for intervals (1), (2) and (3) are shown in Fig. 5(a)–(c) respectively. The switching period in Fig. 5(a)–(c) is $T_{sw} = 2(T_1 + T_2 + T_3 + T_4)$.

Based on Fig. 5(a)–(c), the expressions obtained for $\Delta I_{i,pk.}$ during the intervals (1), (2) and (3) for the 3-phase 3-wire AFEC (a -phase) are shown in Table I. The expressions for $\Delta I_{i,pk.,a}$ in the intervals (4), (5) and (6) can be deduced by the expressions of $\Delta I_{i,pk.,a}$ during the intervals (3), (2) and (1) respectively by interchanging v_b^* by v_c^* and v_c^* by v_b^* , as can be observed from Fig. 4. The expressions of v_a^* , v_b^* , and v_c^* used in Table I are same as (3). The expression of v_{ga} can be found in (4).

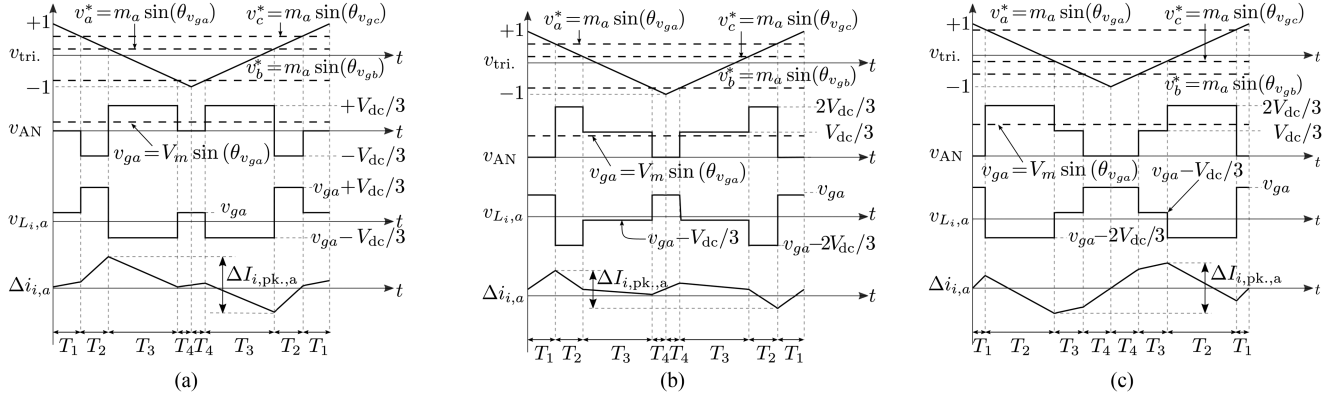


Fig. 5. Plot of v_a^* with $v_{tri.}$, v_{AN} with v_{ga} , $v_{Li,a}$ and $\Delta i_{i,a}$ for one switching period (T_{sw}) for a 3-phase 3-wire AFEC during (a) interval ①, (b) interval ②, and (c) interval ③.

TABLE I
ANALYTICAL EXPRESSIONS OF $\Delta I_{i,pk.,a}$ FOR 3-PHASE 3-WIRE AFEC DURING THE INTERVALS ①, ② AND ③

Interval	Expression of $\frac{\Delta I_{i,pk.,a}}{2}$	Expression of T_x ($x \in 1, 3, 4$)	Expression of $\Delta I_{i,pk.,a}$
①	$\frac{1}{L_i} \left(\frac{V_{dc}}{3} - v_{ga} \right) T_3 - \frac{v_{ga}}{L_i} T_4$	$T_3 = \frac{1}{4f_{sw}} (v_a^* - v_b^*); T_4 = \frac{1}{4f_{sw}} (1 + v_b^*)$	$\frac{1}{2f_{sw} L_i} \left[\frac{V_{dc}}{3} (v_a^* - v_b^*) - v_{ga} (1 + v_a^*) \right]$
②	$\frac{v_{ga}}{L_i} T_1$	$T_1 = \frac{1}{4f_{sw}} (1 - v_a^*)$	$\frac{v_{ga}}{2f_{sw} L_i} (1 - v_a^*)$
③	$\frac{1}{L_i} \left(v_{ga} - \frac{V_{dc}}{3} \right) T_3 + \frac{v_{ga}}{L_i} T_4$	$T_3 = \frac{1}{4f_{sw}} (v_c^* - v_b^*); T_4 = \frac{1}{4f_{sw}} (1 + v_b^*)$	$\frac{1}{2f_{sw} L_i} \left[v_{ga} (1 + v_c^*) + \frac{V_{dc}}{3} (v_b^* - v_c^*) \right]$

III. CALCULATION OF INVERTER-SIDE INDUCTANCE (L_i) FOR THE 3-PHASE AFEC

In Sections II-A and II-B, it is observed that the equations of $\Delta I_{i,pk.,a}$ [(5) for the 3-phase 4-wire AFEC and Table I for the 3-phase 3-wire AFEC] are function of θ_{iia} and L_i for a fixed V_{dc} , m_a , and $V_{g,nom.}$. Therefore, for obtaining the profile of $\Delta I_{i,pk.,a}$ versus θ_{iia} , the value of L_i must be known. The value of L_i is calculated based on the allowable maximum peak-peak ripple current ($\Delta I_{i,pk.,m}$).

A. 4-Wire Configuration

From (5), $\Delta I_{i,pk.,m}$ for the 3-phase 4-wire AFEC occurs at $\theta_{iia} = 0^\circ$, which is mathematically expressed as

$$\Delta I_{i,pk.,m} = \frac{V_{dc}}{4f_{sw} L_i}. \quad (7)$$

For $\Delta I_{i,pk.,m} = 20\%$ of inverter's rated fundamental current peak ($I_{i1,pk.} = 13.88\sqrt{2}$ A), the value of L_i is calculated as 5.07 mH using (7) for the 3-phase 4-wire AFEC. The specifications of the AFEC is shown in Table II.

B. 3-Wire Configuration

For the 3-phase 3-wire AFEC, there are six different expressions for $\Delta I_{i,pk.,a}$ as discussed in Section II-B. A graphical approach is used to determine the value of θ_{iia} at which $\Delta I_{i,pk.,a}$

TABLE II
CONVERTER RATINGS

Parameter	Value
Rated power (S)	10 kVA
Nominal grid voltage ($V_{g,L-L}$)	415 V
DC-link voltage (V_{dc})	800 V
Nominal grid frequency (f_1)	50 Hz
Switching frequency (f_{sw})	10.05 kHz

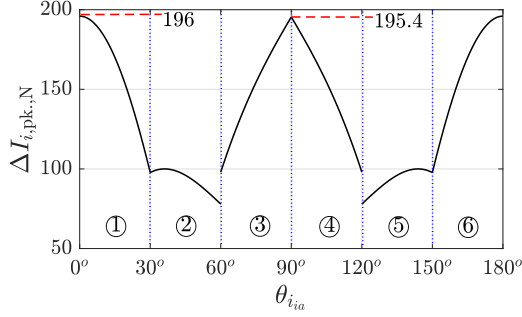
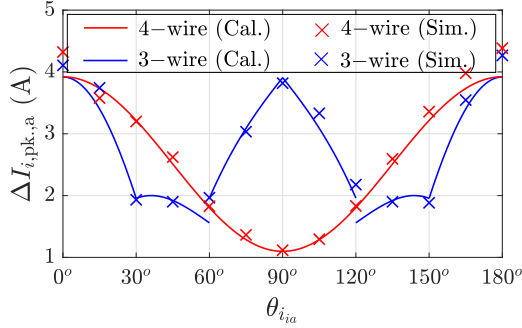
is maximum. $\Delta I_{i,pk.,N}$, the normalized expressions of $\Delta I_{i,pk.,a}$ are obtained in (8)–(10) by multiplying the factor $2f_{sw} L_i$ with the expressions of $\Delta I_{i,pk.,a}$ in Table I. Clearly from (8), (9) and (10), $\Delta I_{i,pk.,N}$ expressions are independent of L_i .

$$\Delta I_{i,pk.,N} = \frac{V_{dc}}{3} (v_a^* - v_b^*) - v_{ga} (1 + v_a^*); \text{Interval ①} \quad (8)$$

$$\Delta I_{i,pk.,N} = v_{ga} (1 - v_a^*); \text{Interval ②} \quad (9)$$

$$\Delta I_{i,pk.,N} = v_{ga} (1 + v_c^*) + \frac{V_{dc}}{3} (v_b^* - v_c^*); \text{Interval ③} \quad (10)$$

The expressions for $\Delta I_{i,pk.,N}$ in the intervals ④, ⑤ and ⑥ can be deduced by the expressions of $\Delta I_{i,pk.,N}$ during the intervals ③, ② and ① respectively by interchanging v_b^* by v_c^* and v_c^* by v_b^* . For a fixed V_{dc} and $v_{g,nom.}$, (8), (9) and (10) are only a function of θ_{iia} . With $V_{dc} = 800$ V and $v_{g,nom.} = 240$ V, the

Fig. 6. Plot of $\Delta I_{i,pk.,N}$ versus θ_{iia} .Fig. 7. Comparison of the variation in $\Delta I_{i,pk.,a}$ with θ_{iia} for the 3-phase 4-wire and the 3-phase 3-wire AFEC.

plot of (8), (9) and (10) w.r.t θ_{iia} is shown in Fig. 6. From Fig. 6, it can be observed that $\Delta I_{i,pk.,N}$ is maximum at $\theta_{iia} = 0^\circ$, hence, $\Delta I_{i,pk.,m}$ for the 3-phase 3-wire AFEC is derived by substituting $\theta_{iia} = 0^\circ$ in the expression of $\Delta I_{i,pk.,a}$ corresponding to interval ① as

$$\Delta I_{i,pk.,m} = \frac{\sqrt{3} m_a V_{dc}}{12 f_{sw} L_i}. \quad (11)$$

For $\Delta I_{i,pk.,m} = 20\%$ of $I_{i1,pk.,}$, the value of L_i is calculated as 2.5 mH using (11) for 3-phase 3-wire AFEC whose specifications are mentioned in Table II.

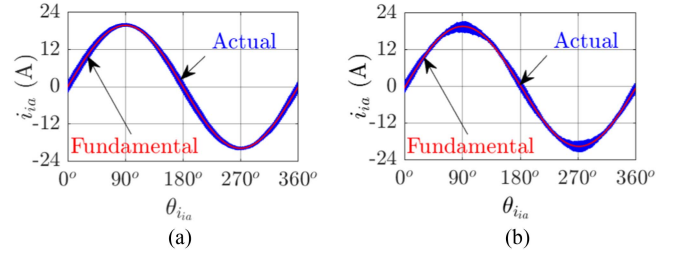
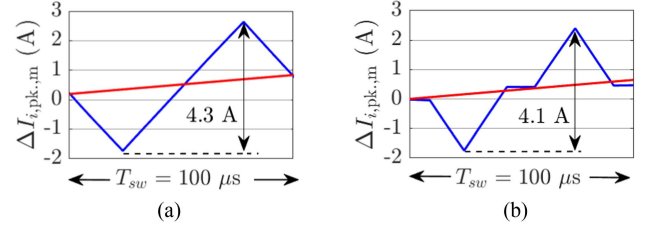
From the above discussions, it is observed that for the same $\Delta I_{i,pk.,m}$, a 3-phase 4-wire AFEC requires twice the value of L_i when compared to a 3-phase 3-wire AFEC.

IV. PROFILE OF $\Delta I_{i,pk.,}$ VERSUS θ_{iia} FOR THE 3-PHASE AFEC

With the known values of L_i for both the AFECs (3-phase 4-wire and 3-phase 3-wire), the equations of $\Delta I_{i,pk.,a}$ [(5) for the 3-phase 4-wire AFEC and Table I for the 3-phase 3-wire AFEC] are now dependent only on θ_{iia} for a fixed V_{dc} , m_a , and $V_{g,nom.}$. Therefore, it is possible to plot $\Delta I_{i,pk.,a}$ for different values of θ_{iia} (0° to 180°) for a fixed V_{dc} , m_a , and $V_{g,nom.}$. Fig. 7 shows the plot of $\Delta I_{i,pk.,a}$ versus θ_{iia} for both the configurations. An RC-damped LCL filter is considered for the analysis and experimental validation. As indicated in Section II, the discussions in the earlier Sections are valid for RC damping as well. The shunt branch in an RC-damped LCL filter has a filter capacitor C_f in parallel with a series combination of

TABLE III
LCL FILTER PARAMETERS

Parameters	Values (4-wire)	Values (3-wire)
L_i	5.07 mH	2.5 mH
L_g	1.25 mH	0.89 mH
C_f	3.6 μ F	3.6 μ F
C_d	5.6 μ F	5.6 μ F
R_d	23 Ω	18.5 Ω

Fig. 8. Simulation result showing the inverter side current (actual and the fundamental component) for $\theta_{iia} = 0^\circ$ to 360° for the (a) 3-phase 4-wire AFEC and (b) 3-phase 3-wire AFEC.Fig. 9. Simulation result showing $\Delta I_{i,pk.,m}$ for the (a) 3-phase 4-wire AFEC and (b) 3-phase 3-wire AFEC.

damping branch capacitor C_d and damping resistor R_d . Fig. 7 is obtained using (5) with $L_i = 5.07$ mH for the 3-phase 4-wire AFEC and using equations in Table I with $L_i = 2.5$ mH for the 3-phase 3-wire AFEC. The complete filter parameters are listed in Table III for both configurations. The obtained profile of $\Delta I_{i,pk.,a}$ versus θ_{iia} is validated with simulation results by implementing the 10kVA AFEC (4-wire and 3-wire) in Matlab Simulink environment. It can be observed that the analytical and simulation results match well with each other for both configurations.

Fig. 8(a) and (b) show the simulation results of the a-phase inverter-side current (i_{ia}) for the 3-phase 4-wire and the 3-phase 3-wire AFEC respectively using the output LCL filter of Table III for one complete fundamental cycle. As per the analysis from Fig. 7, the value of ripple current in the 3-phase 4-wire AFEC is highest at $\theta_{iia} = 0^\circ$ and lowest at $\theta_{iia} = 90^\circ$, whereas, the value of ripple current in the 3-phase 3-wire AFEC is highest at the $\theta_{iia} = 0^\circ$ with the ripple value being almost same at $\theta_{iia} = 0^\circ$ and $\theta_{iia} = 90^\circ$. These inferences can also be observed from Fig. 8(a) for the 4-wire case and from Fig. 8(b) for a 3-wire case. Fig. 9(a) and (b) show the maximum peak-peak ripple current ($\Delta I_{i,pk.,m}$) for the 4-wire and 3-wire AFEC respectively,

which occurs at $\theta_{i_{ia}} = 0^\circ$. For the respective values of L_i as in Table III, the value of $\Delta I_{i,pk,m}$ is expected to be 3.94 A. From Fig. 9(a) and (b) it is seen that the results show a close match.

V. PROFILE OF $\Delta I_{i,pk}$ VERSUS θ_{i_i} FOR A 3-PHASE GCI AS DGAS

Distributed sources make use of GCIs for injecting the generated power into the grid. The generated power of the distributed source feeds the dc-bus of a GCI shown in Fig. 1. The GCI acts as an interface between the distributed source and the grid, injecting the real power into the grid. The grid current is maintained in phase with the grid voltage as in an AFEC. However, the direction of the grid current is into the PCC in this case. So, the expression for the voltage drop across L_i for a -phase is

$$v_{L_i,a} \approx -(v_{ga} - v_{AN}). \quad (12)$$

Therefore, from (12) and (2), it can be clearly inferred that the waveforms of $v_{L_i,a}$ and $\Delta i_{i,a}$ in case of GCIs for distributed generation applications (for both 3-wire and 4-wire configurations) are the same as that of an AFEC except that they are inverted about the x -axis due to opposite current directions. Hence the profile of $\Delta I_{i,pk,a}$ versus $\theta_{i_{ia}}$ for a GCI used in DGAs is the same as shown in Fig. 7. The observations that are drawn from Fig. 7 in Section IV applies for this case also.

Since $\Delta I_{i,pk,a}$ versus $\theta_{i_{ia}}$ for 4-wire and 3-wire GCI as DGAs are the same as that of an AFEC of the 4-wire and 3-wire configurations respectively, the simulation results for GCI as DGAs are not shown here.

VI. PROFILE OF $\Delta I_{i,pk}$ VERSUS θ_{i_i} FOR A 3-PHASE STATCOM

A STATCOM is used at the utility end to meet the reactive power demand of the load in order to ensure a unity power-factor operation. From Fig. 1, considering the current direction into the inverter, the inverter current must lead the grid voltage by 90° to meet the load reactive power demand.

A. 4-Wire Configuration

In a STATCOM, since $\theta_{v_{ga}} \approx \theta_{i_{ia}} - 90^\circ$, the equation of $\Delta I_{i,pk,a}$ for a 3-phase 4-wire STATCOM can be obtained by modifying (5) as

$$\Delta I_{i,pk,a} = \frac{1}{2f_{sw}L_i} \left[\frac{V_{dc}}{2} - \sqrt{2}m_a V_{g,nom.} \sin^2(\theta_{i_{ia}} - 90^\circ) \right]. \quad (13)$$

B. 3-Wire Configuration

The expressions of $\Delta I_{i,pk,a}$ in the case of a 3-phase 3-wire STATCOM can be obtained by understanding the phasor relation between the a -phase current of the STATCOM with the inverter reference voltages as shown in Fig. 10 and comparing it with Fig. 4. From Figs. 4 and 10, the expressions of $\Delta i_{i,pk,a}$ for the 3-phase 3-wire STATCOM for $90^\circ \leq \theta_{i_{ia}} \leq 120^\circ$, $120^\circ \leq \theta_{i_{ia}} \leq 150^\circ$ and $150^\circ \leq \theta_{i_{ia}} \leq 180^\circ$ are same as $\Delta i_{i,pk,a}$ in intervals ①, ② and ③ respectively of Table I with $v_{ga} = \sqrt{2}V_{g,nom.} \sin(\theta_{i_{ia}} - 90^\circ)$, $v_a^* =$

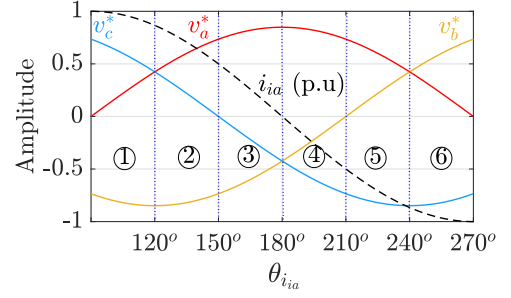


Fig. 10. Inverter reference voltages (v_a^* , v_b^* and v_c^*) and the inverter side fundamental current (i_{ia}) in p.u. of a 3-phase 3-wire STATCOM.

$m_a \sin(\theta_{i_{ia}} - 90^\circ)$, $v_b^* = m_a \sin(\theta_{i_{ia}} - 90^\circ - 120^\circ)$ and $v_c^* = m_a \sin(\theta_{i_{ia}} - 90^\circ + 120^\circ)$. For example, for $90^\circ \leq \theta_{i_{ia}} \leq 120^\circ$ the expression of $\Delta I_{i,pk,a}$ can be obtained as

$$\Delta I_{i,pk,a} = \frac{1}{2f_{sw}L_i} \left[\frac{V_{dc}}{3} m_a \{ \sin(\theta_{i_{ia}} - 90^\circ) - \sin(\theta_{i_{ia}} - 90^\circ - 120^\circ) \} - \sqrt{2} V_{g,nom.} \sin(\theta_{i_{ia}} - 90^\circ) \{ 1 + m_a \sin(\theta_{i_{ia}} - 90^\circ) \} \right]. \quad (14)$$

Similarly, the expressions of $\Delta i_{i,pk,a}$ for $0^\circ \leq \theta_{i_{ia}} \leq 30^\circ$, $30^\circ \leq \theta_{i_{ia}} \leq 60^\circ$ and $60^\circ \leq \theta_{i_{ia}} \leq 90^\circ$ are same as $\Delta i_{i,pk,a}$ in intervals ④, ⑤ and ⑥ respectively as discussed in Section II-B with $v_{ga} = V_m \sin(\theta_{i_{ia}} + 180^\circ - 90^\circ)$, $v_a^* = m_a \sin(\theta_{i_{ia}} + 180^\circ - 90^\circ)$, $v_b^* = m_a \sin(\theta_{i_{ia}} + 180^\circ - 90^\circ - 120^\circ)$, and $v_c^* = m_a \sin(\theta_{i_{ia}} + 180^\circ - 90^\circ + 120^\circ)$.

Following a similar approach as discussed in Section III, it can be shown that the maximum peak-peak ripple current ($\Delta I_{i,pk,m}$) occurs at $\theta_{i_{ia}} = 90^\circ$ for the 10kVA 3-phase 4-wire and 3-phase 3-wire STATCOM with $V_{dc} = 800$ V, $v_{g,nom.} = 240$ V and $f_{sw} = 10$ kHz. For the 3-phase 4-wire STATCOM, by substituting $\theta_{i_{ia}} = 90^\circ$ in (13), the expression of $\Delta I_{i,pk,m}$ can be obtained as

$$\Delta I_{i,pk,m} = \frac{V_{dc}}{4f_{sw}L_i}. \quad (15)$$

The expression of $\Delta I_{i,pk,m}$ for the 3-phase 3-wire STATCOM is derived as

$$\Delta I_{i,pk,m} = \frac{\sqrt{3}m_a V_{dc}}{12f_{sw}L_i} \quad (16)$$

by substituting $\theta_{i_{ia}} = 90^\circ$ in (14).

For the converter ratings in Table II, the value of L_i for $\Delta I_{i,pk,m} = 20\%$ of $I_{i1,pk}$ for the 3-phase 4-wire and 3-phase 3-wire STATCOM are calculated as 5.07 mH and 2.5 mH respectively using (15) and (16). Fig. 11 shows the comparison between the analytical and simulated $\Delta I_{i,pk,a}$ versus $\theta_{i_{ia}}$ between 3-phase 4-wire and 3-phase 3-wire STATCOM with the LCL filter of Table III. It is observed that the analytical and simulation results match well with each other for both configurations.

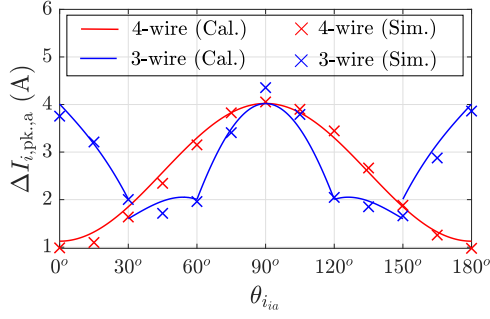


Fig. 11. Comparison of the variation in $\Delta I_{i,pk.,a}$ with θ_{iia} for the 3-phase 4-wire and the 3-phase 3-wire STATCOM.

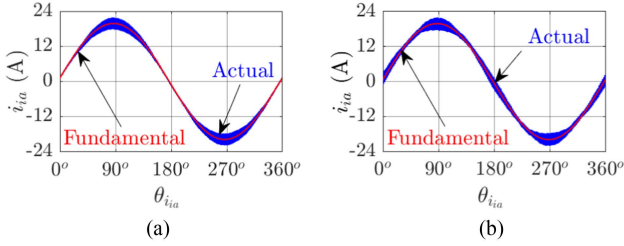


Fig. 12. Simulation result showing the inverter side current (actual and the fundamental component) for $\theta_{iia} = 0^\circ$ to 360° for (a) 3-phase 4-wire STATCOM and (b) 3-phase 3-wire STATCOM.

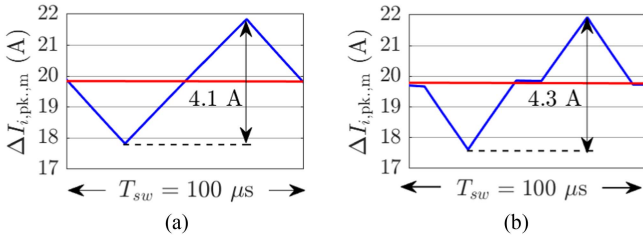


Fig. 13. Simulation result showing $\Delta I_{i,pk.,m}$ for (a) 3-phase 4-wire STATCOM and (b) 3-phase 3-wire STATCOM.

Fig. 12(a) and (b) show the simulation results of the inverter-side current for the 3-phase 4-wire and 3-phase 3-wire STATCOM respectively using the output LCL filter in Table III for one complete fundamental cycle. The ripple pattern in the currents is as per the earlier analysis as shown in Fig. 11. For the 3-phase 4-wire configuration the current ripple is maximum at $\theta_{iia} = 90^\circ$ and minimum at $\theta_{iia} = 0^\circ$ [Fig. 12(a)], whereas, for the 3-phase 3-wire configuration the ripple current value is highest at $\theta_{iia} = 90^\circ$ with almost same ripple value at $\theta_{iia} = 90^\circ$ and $\theta_{iia} = 0^\circ$ [Fig. 12(b)]. Fig. 13(a) and (b) show $\Delta I_{i,pk.,m}$ for the 4-wire and 3-wire STATCOM respectively. From Fig. 13(a) and (b) the obtained $\Delta I_{i,pk.,m}$ is close to the expected value of 3.94 A.

VII. EFFECT OF L_i ON RESONANCE FREQUENCY AND SYSTEM STABILITY IN LCL FILTERS WITH PASSIVE DAMPING

In this work, L_i is selected based on $\Delta I_{i,pk.,m}$ which can range from 10% to 30%. The other parameters of the filter are

designed using the method proposed in [18]. Different values of $\Delta I_{i,pk.,m}$ result in different values of L_i , leading to variations in the remaining filter parameters. Therefore, it is necessary to analyze the effects of selecting L_i based on $\Delta I_{i,pk.,m}$ on the closed-loop stability and resonance damping.

Table IV shows the various values of L_i obtained by considering different $\Delta I_{i,pk.,m}$ (between 10% to 30%) for the 3- and 4-wire GCI. The other parameters of the filter, also shown in Table IV, vary according to the different values of L_i . The filter design method in [18] offers a good resonance damping with the use of a high damping resistor (R_d) while maintaining a low damping loss. This also ensures proper stability of the GCI.

The grid-side current control is employed for control of the GCI. The plant transfer (TF) used for the implementation of the closed-loop control is

$$G_{RC-LCL}(s) = \frac{I_g(s)}{V_i(s)} = \frac{1 + sR_dC_d}{sL_T + s^2R_dC_dL_T + s^3L_iL_gC_T + s^4R_dL_iL_gC_fC_d}. \quad (17)$$

where $L_T = L_i + L_g$ and $C_T = C_f + C_d$. The compensated open-loop transfer function (TF) of the current control in the z -domain is obtained as

$$G_{ol,comp}(z) = z^{-1}G_{ol,uncomp}(z)G_{PI}(z). \quad (18)$$

In (18), $G_{PI}(z)$ and $G_{ol,uncomp}(z)$ are the z domain TF of the PI controller and $G_{RC-LCL}(s)$ respectively. $G_{ol,uncomp}(z)$ is obtained by zero-order hold (ZOH) discretization of $G_{RC-LCL}(s)$, to account for $0.5T_s$ PWM delay (T_s being the sampling time) [19], [20]. The additional T_s digital processing delay is accounted by z^{-1} in (18) [19], [21]. For the controller implementation, the sampling frequency (f_s) is kept equal to the switching frequency (f_{sw}).

The selection of the controller gain is done using the method outlined in [22], [23] which is based on a desired phase margin (PM) at the design stage. The controller gains (K_p and K_i) are shown in Table IV for a desired PM of 60° . Fig. 14 shows the bode plots of $G_{ol,comp}(z)$ of the GCI controls for the different filters listed in Table IV. Fig. 15 shows the pole-zero mapping of the closed-loop transfer function

$$G_{cl}(z) = \frac{G_{ol,comp}(z)}{1 + G_{ol,comp}(z)}. \quad (19)$$

The obtained PM and gain margin (GM) of $G_{ol,comp}(z)$ is listed in Table IV. It can be observed that the GM of the $G_{ol,comp}(z)$ reduces with decreasing L_i (increasing $\Delta I_{i,pk.,m}$). A similar trend is seen in the PM too. The pole-zero plots in Fig. 15 also support the above observation. From Fig. 15, it is seen that the poles move towards the unity circle for smaller values of L_i (higher $\Delta I_{i,pk.,m}$).

It can also be seen from Table IV that the obtained PM is lower than the desired PM of 60° . This discrepancy arises because the approximate model of the plant TF used for the controller design outlined in [22], [23] does not match the actual plant TF well at high frequencies. This mismatch is due to the dominance of the shunt branch at low frequencies for lower values of L_i .

TABLE IV
PARAMETERS OF THE RC -DAMPED LCL FILTER FOR DIFFERENT $\Delta I_{i,\text{pk},m}$, CORRESPONDING RESONANCE FREQUENCY AND THE CLOSED-LOOP CURRENT CONTROLLER GAINS ALONG WITH THE OBTAINED GAIN MARGIN (GM) AND PHASE MARGIN (PM)

$\Delta I_{i,\text{pk},m}$ (% of $I_{i1,\text{pk}}$)	Configuration	L_i (mH)	L_g (mH)	C_d (μF)	C_f (μF)	R_d (Ω)	f_r (kHz)	K_p	K_I	PM (degree)	GM (dB)
10	3-wire	5.00	0.46	5.6	3.6	15.0	3.06	19.1	2218	56.0	6.5
	4-wire	10.2	0.64	5.6	3.6	17.7	2.56	37.8	4398	55.0	5.3
15	3-wire	3.34	0.66	5.6	3.6	17.0	2.68	13.9	1622	55.6	5.6
	4-wire	6.80	0.93	5.6	3.6	20.7	2.20	27.0	3136	53.5	4.2
20	3-wire	2.50	0.89	5.6	3.6	18.5	2.48	11.7	1366	55.0	5.0
	4-wire	5.09	1.25	5.6	3.6	22.7	2.00	22.0	2569	52.0	3.4
25	3-wire	2.00	1.07	5.6	3.6	19.0	2.38	10.7	1246	54.5	4.8
	4-wire	4.07	1.50	5.6	3.6	24.0	1.89	19.5	2269	51.0	3.0
30	3-wire	1.67	1.28	5.6	3.6	19.5	2.34	10.3	1196	54.4	4.6
	4-wire	3.39	1.80	5.6	3.6	25.0	1.83	18.0	2113	50.4	2.8

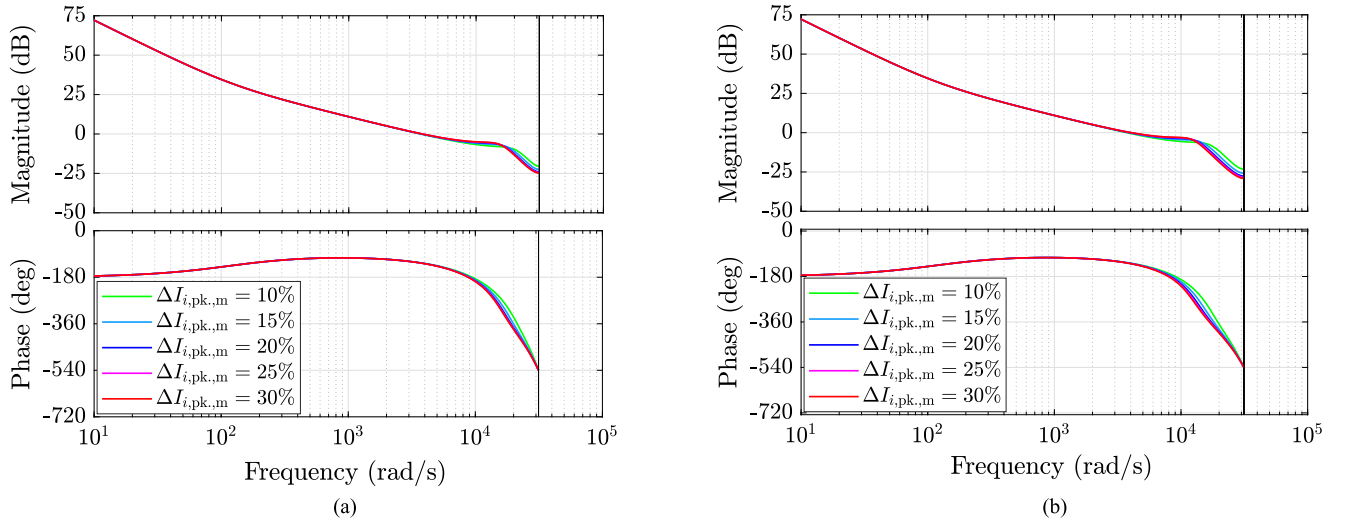


Fig. 14. Bode plots of $G_{ol,\text{comp}}(z)$ for the closed-loop control of the GCI using the filter parameters listed in Table IV: (a) 3-wire configuration and (b) 4-wire configuration.

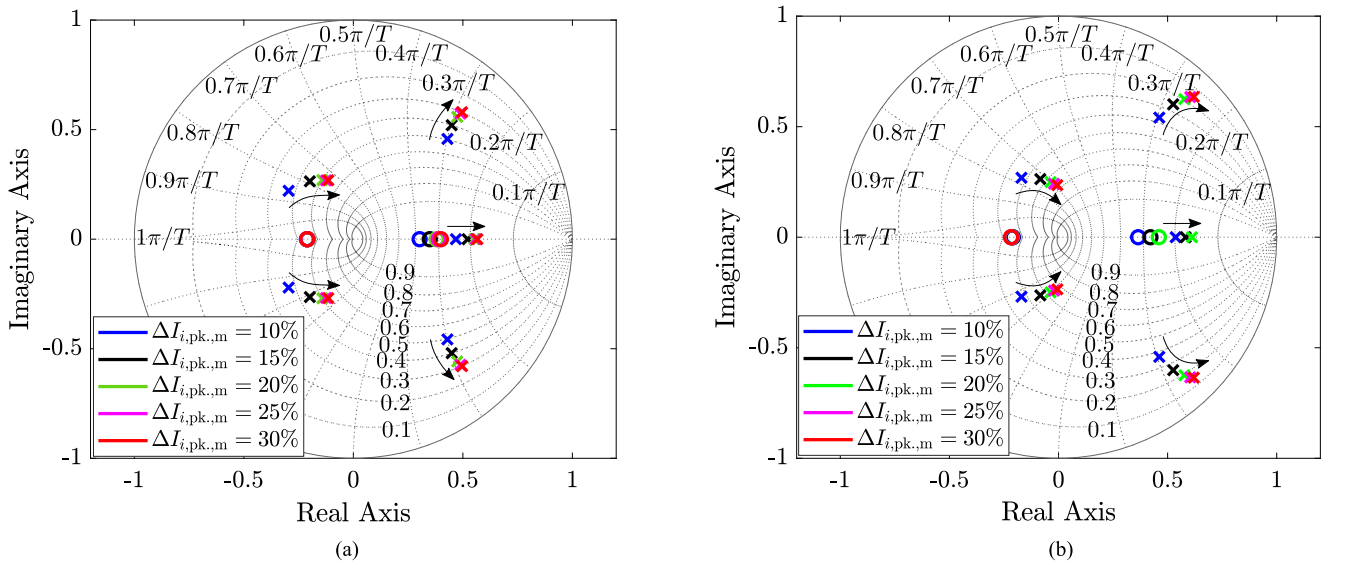


Fig. 15. Pole-zero plot of $G_{cl}(z)$ with the filter and PI controller parameters filter of Table IV for (a) 3-wire configuration and (b) 4-wire configuration.

TABLE V
PARAMETERS OF LCL FILTER FOR DIFFERENT $\Delta I_{i,pk.,m}$

$\Delta I_{i,pk.,m}$ (% of $I_{i1,pk.}$)	Type	L_i (mH)	C_f (μ F)	L_g (mH)	f_r (Hz)	$f_r > \frac{f_s}{6}$
10	3	5	9.2	0.183	3.95	Yes
	4	10.2	9.2	0.25	3.38	Yes
15	3	3.34	9.2	0.26	3.38	Yes
	4	6.80	9.2	0.36	2.85	Yes
20	3	2.50	9.2	0.34	3.04	Yes
	4	5.09	9.2	0.47	2.53	Yes
25	3	2.00	9.2	0.42	2.82	Yes
	4	4.07	9.2	0.58	2.33	Yes
30	3	1.67	9.2	0.50	2.68	Yes
	4	3.39	9.2	0.7	2.19	Yes

From Table IV it can also be observed that the 3-wire GCI with an output RC -damped LCL filter has higher GM and PM when compared to the 4-wire configuration for the same $\Delta I_{i,pk.,m}$. Fig. 15 supports this observation, showing that the poles are much closer to the unity circle in the case of 4-wire GCI compared to the 3-wire GCI for a fixed $\Delta I_{i,pk.,m}$.

Furthermore, the bode plots of Fig. 14 suggest that the resonance damping of the RC -damped LCL filter is unaffected by the choice of L_i when the other parameters of the filter are designed using [18]. A similar result can be observed for the R -damped LCL filter when the remaining filter parameters are designed using the method proposed in [12].

VIII. EFFECT OF L_i ON SYSTEM STABILITY IN LCL FILTERS WITHOUT PASSIVE DAMPING

Section VII gives a detailed analysis showing that the closed-loop control is stable for different values of L_i when the parameters of the passive damped LCL filter (R - and RC -damping) are selected as suggested in [12], [18]. This section investigates the stability of the LCL filter for such a choice of L_i without passive damping.

Table V shows the parameters of the LCL filter for both 3- and 4-wire configurations for different values of $\Delta I_{i,pk.,m}$. The filter capacitor (C_f) is selected based on the 5% reactive power criteria and the grid-side inductor (L_g) is determined such that the filter with the selected L_i and C_f achieves the desired attenuation at the switching frequency as mandated by the IEEE-1547 standard.

Three distinct regions of LCL filter resonance frequency (f_r) separated by one-sixth of the sampling frequency $f_s/6$ are identified in [24]. The frequency $f_r = f_s/6$ is termed as the critical frequency, $f_{critical}$. Therefore, the three resonant frequency regions are: (i) low resonant frequency region where $f_r < f_{critical}$, (ii) critical resonant frequency, and (iii) high resonant frequency region where $f_r > f_{critical}$ [19].

Table V shows the filter parameters and the resonance frequency of the LCL filters. In this work, the sampling frequency is equal to the switching frequency which is 10 kHz. Hence, $f_{critical} = 10000/6 = 1.67$ kHz. From Table V, it is clear that the resonance frequency of the filters lies in the high resonant frequency region. Therefore, the grid-side current control of the GCI with the LCL filters listed in Table V will be stable with an

appropriate choice of PI controller parameters without the need for active damping [24].

The filters (for both 3- and 4-wire) with $\Delta I_{i,pk.,m} = 30\%$ of $I_{i1,pk.}$ have f_r closest to $f_{critical}$. Therefore, the closed-loop poles of the current control with the above two filters are obtained to support the stability statement made in the above paragraph.

The plant TF for implementing the closed-loop current control with grid-current feedback is

$$G_{LCL}(s) = \frac{I_g(s)}{V_i(s)} = \frac{1}{s(L_i + L_g) + s^3 L_i L_g C_f}. \quad (20)$$

The closed-loop poles are obtained by solving (21) by following the discretization technique discussed in Section VII.

$$1 + K_p z^{-1} G_{LCL}(z) = 0 \quad (21)$$

Fig. 16 shows the root loci of the closed-loop poles of (21). From Fig. 16, it is observed that the closed-loop control system is stable with appropriate controller parameters without the need for active damping even for the worst case scenarios (filters having f_r closest to $f_{critical}$).

IX. CHOICE OF INDUCTOR CORE MATERIAL

The choice of the core material is influenced by various factors such as the switching frequency, core loss, cost, size, and weight of the core. Below are considerations for selecting the core type for the inverter-side inductor, L_i :

- 1) **High $B_{sat.}$:** The core should have a high saturation flux density ($B_{sat.}$) to ensure a smaller core size. The allowable peak flux density ($B_{max.} = 90\% B_{sat.}$) of the core is given by

$$B_{max.} = \frac{L_i I_{pk.}}{N A_e} \quad (22)$$

where $L_i = 2.5$ mH, and for the converter specifications in Table II, $I_{pk.} = I_{i1,pk.} + \frac{\Delta I_{i,pk.,m}}{2} = (13.88\sqrt{2}) + (\frac{3.94}{2}) = 21.6$ A. In (22), N is the number of turns of the inductor while A_e is the cross-section area of the core used, as depicted in Fig. 17. It is evident from (22) that a higher value of $B_{sat.}$ will result in a smaller core dimension (A_e) for a given value of $I_{pk.}$.

- 2) **Higher Bandwidth of μ :** For the 3-wire AFEC and the 3-wire STATCOM discussed in Sections III and VI respectively, it may be noted that the value of L_i is based on the ripple current requirement ($\Delta I_{i,pk.,m}$). So, the required value of L_i is to be achieved at the switching frequency of $f_{sw} = 10$ kHz. The value of L_i is also important at the fundamental frequency $f_1 = 50$ Hz, for current control. Therefore, L_i should be maintained constant over the fundamental frequency to the switching frequency.

The expression of the inductance for the inductor with the core geometry shown in Fig. 17 is given by

$$L = \frac{N^2}{\frac{l_m - 2l_g}{\mu A_e} + \frac{2l_g}{\mu_o A_e}} \quad (23)$$

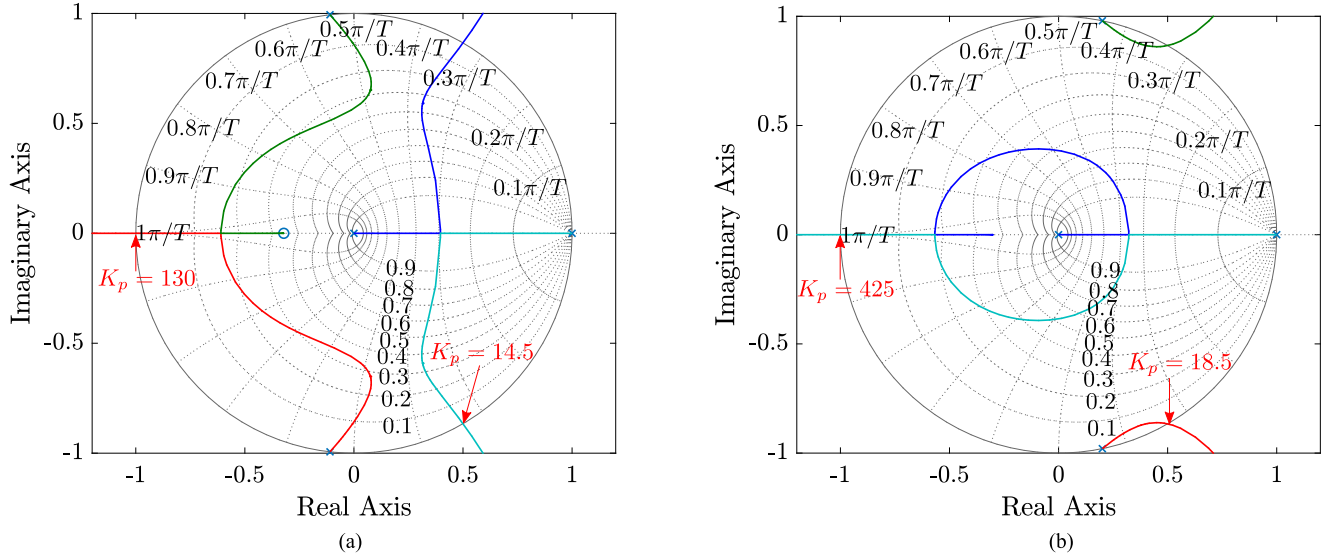


Fig. 16. Root loci of closed-loop poles for (a) 3-wire configuration for $\Delta I_{i,pk,m} = 30\%$ of $I_{i1,pk}$. (b) 4-wire configuration for $\Delta I_{i,pk,m} = 30\%$ of $I_{i1,pk}$.

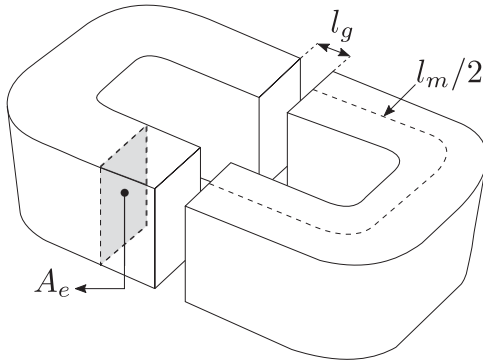


Fig. 17. C-core dimensions.

where μ_o is the permeability of free space and μ is the permeability of the core.

From (23), it can be seen that a reduction in μ would result in a reduction in L . Therefore, a constant μ is desired over the fundamental frequency to the switching frequency to ensure a constant L_i in that range of frequency variation.

- 3) **Minimal Variation of μ with Current:** The permeability of the selected core should be less sensitive to current bias such that a time-varying inductance can be avoided over the sinusoidal variation in current. Typically, the permeability of the core drops at higher values of current, as we approach saturation. A reduced permeability can cause a reduction in the inductance value, leading to a higher-than-expected ripple. It may be noted that in a 3-wire AFEC and STATCOM, this is compounded by the fact that the maximum peak-to-peak ripple current appears at the peak of the fundamental current. The increased ripple current can affect the losses in the inverter switches. Further, a time-varying inductance when used as a differential mode filter in 3-phase systems, can introduce differential-mode (DM) and common-mode (CM) noise [17].

Iron (laminated Silicon steel – 97 % Fe and 3 % Si), amorphous, and ferrite are possible choices of cores that can meet the above requirements. While powdered and nanocrystalline materials exhibit lower losses and reasonably high saturation flux density, the advantages of these cores are outweighed by the cost disadvantage, making them an inappropriate choice for the specifications in Table II.

The high saturation flux density ($B_{sat.}$) of iron and amorphous core proves to be advantageous, considering the peak value of current, $I_{pk.} = 21.6$ A. $B_{sat.}$ for iron cores and amorphous cores ranges from 0.65 T to 1.8 T and 0.5 T to 1.5 T respectively [25]. However, the permeability of iron cores starts to decline at a relatively lower frequency, well below the switching frequency of 10kHz, when compared to the amorphous cores [26]. Therefore, w.r.t maintaining the value of μ up to the switching frequency, amorphous cores are preferred over iron cores [27]. Also, iron cores exhibit a low permeability at lower current values (low initial permeability, μ_i) due to the nature of its B-H curve, resulting in a lower value of inductance at low currents. Ferrites have good linearity over a wide range of frequencies [27] compared to iron and amorphous cores. However, they have a very low $B_{sat.}$ (0.3 T to 0.5 T) [25]. Based on the above properties, an amorphous core emerges as an appropriate choice of core material for the inverter-side inductor L_i .

X. VARIATION OF INDUCTANCE WITH CURRENT IN AMORPHOUS CORE INDUCTORS

Amorphous core-based inductor design is typically done using the inductance factor (A_L) versus magnetizing force (NI) curves provided by the core manufacturer [15]. Fig. 18 shows the A_L versus $NI_{pk.}$ curves for the AMCC63 core. The A_L versus NI curves are provided for different values of core gap length (l_g) along with the corresponding initial A_L values, A_{L0} (values of A_L when NI is very low). The design method in [29] is considered to demonstrate the variation of inductance

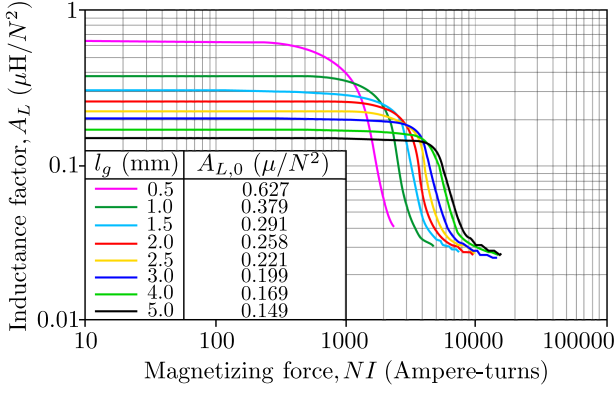


Fig. 18. Inductance factor (A_L) versus magnetizing force curves for AMCC63 core [28].

with current. In [29], after selection of the core size and l_g (by considering the air gap reluctance = $10 \times$ core reluctance), the number of turns (N) for the required inductance L is calculated by

$$L = A_L N^2 \quad (24)$$

where A_L is considered as $A_{L,0}$ corresponding to the calculated l_g .

For the AMCC63 core, following the method in [29], l_g is calculated to be around 0.5 mm ($l_m = 253$ mm and $\mu_r = 3000$). From Fig. 18, $A_{L,0} = 0.627 \mu\text{H}/(\text{turns})^2$ corresponding to $l_g = 0.5$ mm. For $L = L_i = 2.5$ mH and $A_L = A_{L,0} = 0.627 \mu\text{H}/(\text{turns})^2$, N is calculated as 63 turns using (24). With $N = 63$ turns, the value of NI at $I = I_{pk,m} = 24$ A is 1512 Ampere-turns [$I_{pk,m} = 1.1 I_{pk} = (1.1 \times 21.6)$ A, for an additional safety factor to avoid core saturation]. From Fig. 18, the value of A_L at 1512 Ampere-turns is $0.196 \mu\text{H}/(\text{turns})^2$. For this value of A_L , the value of L_i is calculated as 0.77 mH using (24). Clearly, the inductance drops drastically from 2.5 mH to 0.77 mH when the current varies from 0 A to $I_{pk,m} = 24$ A.

From the above discussion, it is clear that the typical approach to amorphous core inductor design as in [29] does not achieve the desired inductance, especially at the peak current. To address this issue, a novel, modified area product method is proposed in the next section for the amorphous core inductor design. The proposed design method ensures that the inductance is much less sensitive to current bias compared to the existing methods.

XI. PROPOSED DESIGN PROCEDURE FOR AMORPHOUS CORE INDUCTOR

A Metglas 2605SA1 (Fe-based amorphous material) AMCC series cut core is used in the design example to fabricate L_i . The magnetic and physical properties of the core are listed in Table VI. Following is the step-by-step design procedure for the amorphous core inductor:

A. Core Size

The core size is selected based on the area product method [15]. The required core area product is calculated as

TABLE VI
MAGNETIC AND PHYSICAL PROPERTIES OF METGLASS AMORPHOUS ALLOY 2605SA1 [16], [30]

Parameters	Values
Saturation flux density, $B_{sat.}$ (T)	1.56
Initial relative permeability	3000-6000
Electrical Resistivity ($\mu\Omega\text{-cm}$)	130
Mass density (g/cm^3)	7.18
Continuous operating temperature ($^\circ\text{C}$)	150
Curie Temperature ($^\circ\text{C}$)	399

$$W_a A_e = \frac{2E10^4}{B_{max.} J_m K_u} = 51.3 \text{ cm}^4 \quad (25)$$

for $B_{max.} = 0.9 B_{sat.}$, K_u (window utilization factor) = 0.4, J_m (current density at $I_{pk,m}$) = $500 \text{ A}/\text{cm}^2$ and E (energy stored in the inductor) = $0.5 L_i I_{pk,m}^2 = 0.72 \text{ J}$ [$I_{pk,m} = 1.1 I_{pk.}$, for additional safety factor to avoid core saturation]. An AMCC63 core [28] having an area product of 55.1 cm^4 [higher than the calculated $W_a A_e$ in (25)] is selected.

B. Number of Turns (N)

In the proposed approach, the required number of turns for the inductor is chosen to meet the inductance required at the peak value of the current. The number of turns to achieve $L_i = 2.5$ mH at $I_{pk,m} = 24$ A can be calculated from

$$L_i = A_{L,pk.} N^2 \quad (26)$$

where $A_{L,pk.}$ is the inductance factor at $I = I_{pk,m}$. With a fourth-order polynomial curve-fitting on the A_L versus NI curves of Fig. 18, the equation of A_L in terms of NI is obtained as

$$A_L = a_4 (NI)^4 + a_3 (NI)^3 + a_2 (NI)^2 + a_1 (NI) + a_0 \quad (27)$$

where a_4 , a_3 , a_2 , a_1 and a_0 are constants obtained from the curve-fitting. The data points of the curves can be extracted using tools such as the Engauge Digitizer.

Using (26) and (27) [at $I = I_{pk,m}$, $A_L = A_{L,pk.}$], the relationship between L_i and N for a fixed $I_{pk,m}$ is obtained as

$$L_i = a_4 I_{pk,m}^4 N^6 + a_3 I_{pk,m}^3 N^5 + a_2 I_{pk,m}^2 N^4 + a_1 I_{pk,m} N^3 + a_0 N^2. \quad (28)$$

The value of N for achieving the required inductance at the desired $I_{pk,m}$ can be obtained by plotting (28) versus N for that particular value of $I_{pk,m}$. The plot of L_i vs. N at $I_{pk,m} = 24$ A for the AMCC63 core is shown in Fig. 19 for 2 mm, 3 mm, 4 mm and 5 mm air gap lengths (l_g). From Fig. 19, the values of N for $L_i = 2.5$ mH at $I_{pk,m} = 24$ A are obtained as 118, 115, 127, and 126 turns for l_g of 2 mm, 3 mm, 4 mm and 5 mm respectively. The number of turns are least in the case of $l_g = 3$ mm. A lower number of turns would result in lower copper loss.

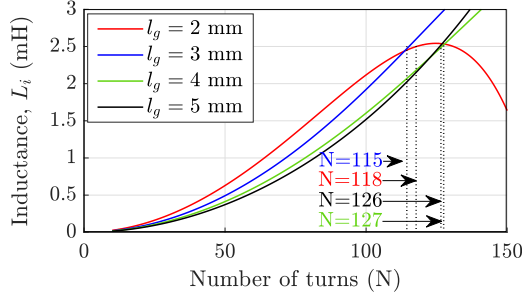
Fig. 19. Plot of L_i versus N for different values of l_g .

TABLE VII
AIR GAP LENGTH VERSUS ΔL FOR $L_i = 2.5$ mH USING AMCC63 CORE

l_g	$L(\text{expected})$	$L(0)$	$L(I_{pk.,m})$	ΔL
2 mm	2.5 mH	3.592 mH	2.5 mH	43.68%
3 mm	2.5 mH	2.631 mH	2.5 mH	5.24%
4 mm	2.5 mH	2.725 mH	2.5 mH	9.00%
5 mm	2.5 mH	2.365 mH	2.5 mH	-5.40%

C. Selection of Air Gap Length (l_g)

The procedure for determining the value of N in order to achieve the desired inductance at the peak current is discussed in Section XI-B for four air gap lengths. At this step, it is required to choose a particular air gap length for the inductor. In this work, it is proposed to select the air gap length such that the designed inductor shows the lowest percentage reduction in its inductance with current bias. The percentage reduction in inductance with current bias can be represented by the metric

$$\Delta L = \frac{L(0) - L(I_{pk.,m})}{L(\text{expected})} \times 100 \% \quad (29)$$

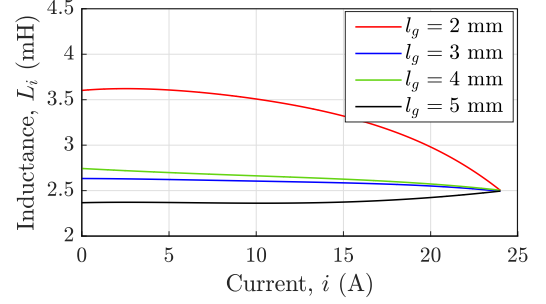
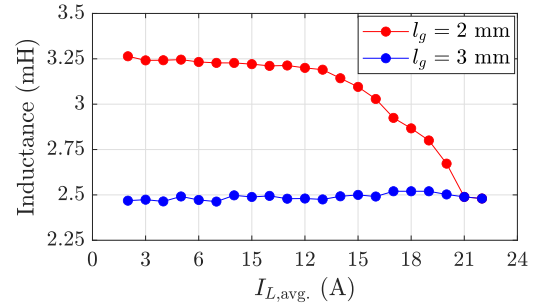
where $L(I_{pk.,m})$ is the inductance at $I_{pk.,m}$, $L(0)$ is the inductance near zero current and $L(\text{expected})$ is the expected value of inductance. A lower value of ΔL would mean less variation of inductance with current bias and is desired.

For example, the value of ΔL for the design using [29] discussed in Section X can be calculated to be 69.2% – indicating a poor design. To determine the l_g which allows for a minimum value of ΔL , Table VII compares the values of ΔL for different l_g considered in Section XI-B. For Table VII, since the value of N is determined using (28), $L(I_{pk.,m}) = 2.5$ mH. The value of $L(0)$ is determined using Fig. 19 [value of N], Fig. 18 [value of initial A_L] and (24). From Table VII, it is clear that ΔL is the minimum for $l_g = 3$ mm.

Although the metric ΔL provides the percentage variation of inductance considering the extreme current values (near zero and peak value), it may not show the exact variation of inductance with current bias. The exact nature of inductance with current bias (i) can be analyzed by

$$L_i = a_4 N^6 i^4 + a_3 N^5 i^3 + a_2 N^4 i^2 + a_1 N^3 i + a_0 N^2 \quad (30)$$

which is obtained using (27) and (24) for a fixed value of N . Fig. 20 shows the plot of L_i versus i (0 A to 24 A) for the

Fig. 20. Plot of L_i versus current (i) for different values of l_g .Fig. 21. Measured inductance at different currents for the 2.5 mH amorphous-core inductor with (i) $l_g = 2$ mm, $N = 118$ turns and (ii) $l_g = 3$ mm, $N = 115$ turns.

AMCC63 core for $l_g = 2$ mm with $N = 118$ turns, $l_g = 3$ mm with $N = 115$ turns, $l_g = 4$ mm with $N = 127$ turns, and $l_g = 5$ mm with $N = 126$ turns. From Fig. 20, it is clear that the variation in L_i is minimal for $l_g = 3$ mm with $N = 115$ turns. Therefore, L_i is fabricated with $N = 115$ turns and $l_g = 3$ mm in the hardware. Also, from Fig. 19, for $l_g = 3$ mm, the number of turns required to achieve $L_i = 2.5$ mH is minimum ($N = 115$ turns) which would result in the lowest copper loss.

The value of l_g is selected such that the designed inductor has a minimum variation in inductance from zero current to peak current. In order to validate the proposed method of selection of l_g , two 2.5 mH inductors were fabricated using the AMCC63 core – (i) $l_g = 2$ mm with $N = 118$ turns and (ii) $l_g = 3$ mm with $N = 115$ turns. The practical variation of inductance with current is plotted for both the inductors in Fig. 21. Fig. 21 is plotted by obtaining the inductance values at different currents (inductor average current) by capturing the voltage and current waveforms of the inductor when used in a buck converter configuration with a switching frequency of 10 kHz. The variation of measured inductance in Fig. 21 shows a similar pattern as in Fig. 20. However, in hardware, the deviation in inductance from the expected value is low at lower currents compared to the analytical results which is shown in Fig. 20.

The metric ΔL is further used to compare the proposed amorphous core inductor design with the existing methods. This metric is useful for assessing the percentage variation in inductance with the current bias for inductors that use a non-linear A_L versus NI characteristic core.

Table VIII compares ΔL of the designed inductor with those obtained with the existing methods. It can be seen that the

TABLE VIII
COMPARISON OF ΔL BETWEEN THE EXISTING METHODS AND THE PROPOSED DESIGN APPROACH

Method	$L(\text{expected})$	$L(0)$	$L(I_{pk,m})$	ΔL
[17]	0.2 mH	0.2 mH	0.09 mH	55%
[11]	0.58 mH	0.87 mH	0.58 mH	50%
[15]	2.761 mH	2.761 mH	2.188 mH	20.75%
[16]	0.15 mH	0.15 mH	0.125 mH	16.67%
[30]	0.4 mH	0.375 mH	0.320 mH	13.75%
Proposed	2.5 mH	2.631 mH	2.5 mH	5.24%

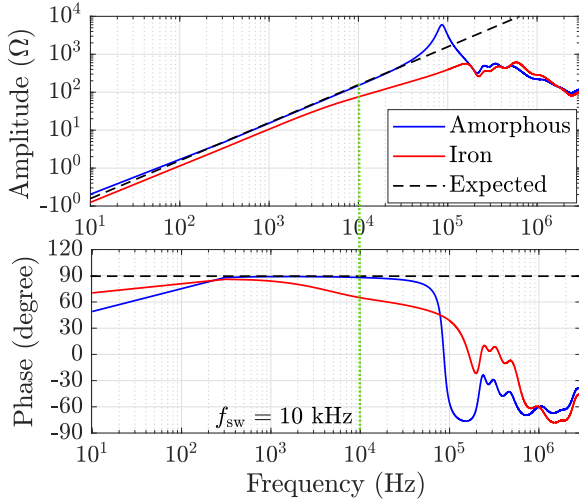


Fig. 22. Comparison of the impedance (amplitude and phase) versus frequency characteristics of the developed 2.5 mH amorphous-core inductor and a 2.5 mH iron-core inductor.

proposed method yields a minimum ΔL , indicating a very stable inductance compared to the existing methods.

XII. COMPARISON OF IRON CORE INDUCTOR WITH AMORPHOUS CORE INDUCTOR

The 2.5 mH amorphous core inductor ($N = 115$ turns and $l_g = 3$ mm) was compared with a 2.5 mH iron core inductor for understanding the superiority of amorphous core in terms of offering a better permeability versus frequency of operation and permeability versus current bias characteristics.

A. Frequency Response

A plot of the impedance (magnitude and phase) versus frequency is obtained in Fig. 22 for the designed iron- and amorphous-core inductors. From Fig. 22, it can be observed that the amorphous-core inductor offers a linear impedance for a wide frequency range when compared to the iron-core inductor. In fact, the impedance ($\bar{Z}$) of the iron-core inductor is observed to be $75.4 \angle 64.8^\circ \Omega$ at 10 kHz against an expected value of $157 \angle 90^\circ \Omega$. The same for the amorphous-core inductor is $156 \angle 88^\circ \Omega$.

With $\bar{Z} = 75.4 \angle 64.8^\circ \Omega$ for the iron-core inductor at 10 kHz and assuming the equivalent model to be comprised of series R and L , the value of inductance and time constant

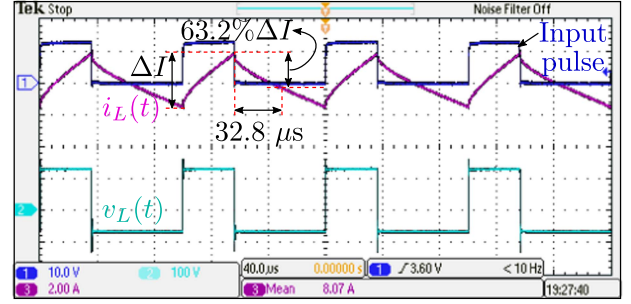


Fig. 23. Voltage ($v_L(t)$) and current $i_L(t)$ of the 2.5 mH iron-core inductor in a buck converter operation.

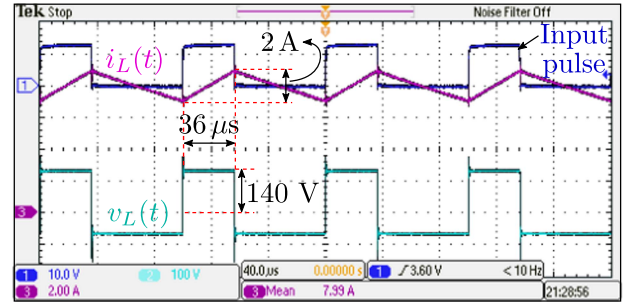


Fig. 24. Voltage ($v_L(t)$) and current $i_L(t)$ of the 2.5 mH amorphous-core inductor in a buck converter operation.

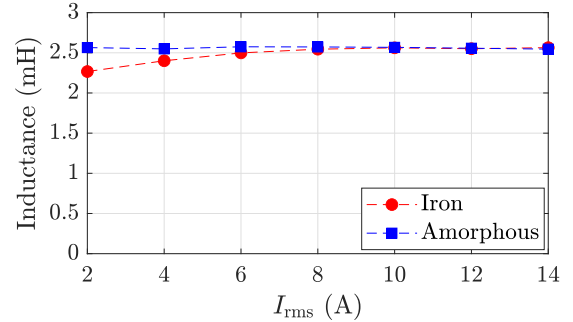


Fig. 25. Variation of inductance with the current bias for the iron- and amorphous-core inductor at 50Hz.

(τ) is calculated to be 1.085 mH and 33.8 μ s respectively. The assumed model is validated by obtaining the experimental value of τ by measuring the current through the iron core inductor when operated in a 10kHz buck converter as shown in Fig. 23. From Fig. 23 it is observed that the experimental value of τ is 32.8 μ s. Therefore, the previously assumed model of the inductor at 10kHz is valid, hence, the inductance of the iron-core inductor is ascertained to be 1.085 mH at 10 kHz frequency against the expected value of 2.5 mH. The reduction in the inductance for the silicon steel-based iron core inductor at 10 kHz is attributed to the decreasing permeability of the material with frequency [31].

From Fig. 24, the measured value of inductance of the amorphous-core inductor at 10 kHz is

$$L = \frac{V_L T_{on}}{\Delta I_L} = 2.52 \text{ mH}. \quad (31)$$

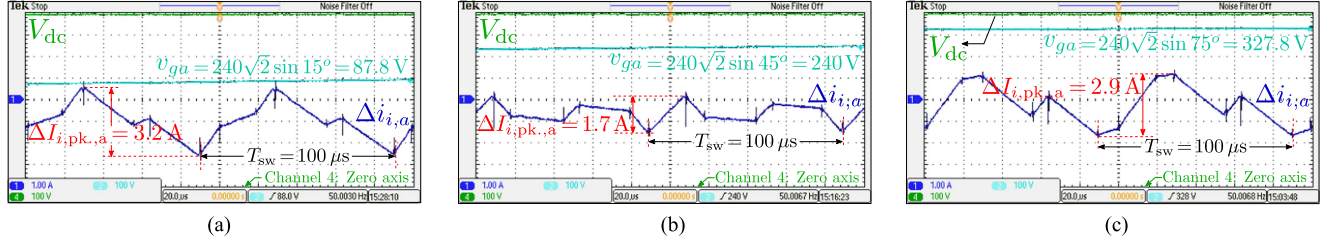


Fig. 26. Experimental result showing $\Delta i_{i,a}$ of the 3-phase 3-wire AFEC during (a) interval ①, (b) interval ②, and (c) interval ③.

The value of inductance at 10 kHz for the amorphous-core inductor from Fig. 22 is obtained as 2.48 mH which is almost the same as the one calculated in (31). The inductance of the designed amorphous-core inductor is very close to the expected value of 2.5 mH at 10 kHz.

B. Inductance Versus Current Bias

Fig. 25 shows the variation of inductance with the current bias for the iron- and amorphous-core inductor at 50Hz operation. The inductance is estimated by measuring the voltage across the inductor for different values of current of frequency 50Hz. It can be observed from Fig. 25 that the iron-core inductor shows lower values of inductance (below 2.5 mH) at lower currents which was mentioned earlier in Section IX. The lower inductance values at low currents can be attributed to the lower initial permeability of the iron core [25]. On the other hand, due to the proposed design method, the inductance of the amorphous-core inductor remains almost constant with the current bias.

XIII. EXPERIMENTAL VALIDATION

In Sections III and VI it is shown that for $\Delta I_{i,pk.,m} = 3.92$ A the value of L_i is 5.07 mH for the 4-wire GCIs (AFEC and STATCOM), whereas the same is only 2.5 mH for the 3-wire GCIs (AFEC and STATCOM). Since the 4-wire configuration requires twice the value of L_i when compared to the 3-wire configuration, only the 3-wire configuration (3-wire AFEC and 3-wire STATCOM) is implemented in the hardware. The 4-wire configuration also requires a larger value of inductance for the grid-side inductor as compared to the 3-wire configuration.

A. Profile of $\Delta I_{i,pk.}$ Versus θ_{iia} for the 3-Phase 3-Wire AFEC and 3-Phase 3-Wire STATCOM

The inverter-side current ripple analysis is carried out based on the assumptions (1), (2), and (3). For the 3-phase GCIs (AFEC, STATCOM, and GCI as DGAs), the assumptions (1), (2) and (3) become explicit for $i_g = 0$. For $i_g = 0$, $v_{Cf,a} = v_{ga}$. Therefore, the proposed ripple analysis can be validated without any current injection. The GCI controls are set to operate the inverter as AFEC and STATCOM with $V_{dc} = 800$ V and $f_{sw} = 10$ kHz, connected to a 415 V (line-line) 50 Hz grid. A grid-side current controller is implemented and the reference current is set to zero to achieve a zero grid-current operation. Fig. 26(a)–(c) show the a-phase inverter ripple current ($\Delta i_{i,a}$) of the 3-phase 3-wire AFEC in the intervals ① (at $\theta_{iia} = 15^\circ$), ② (at $\theta_{iia} = 45^\circ$), and

③ (at $\theta_{iia} = 75^\circ$) respectively. The range of θ_{iia} for intervals ①–③ is shown in Fig. 4 which is discussed in Section II-B. From Fig. 26(a)–(c) it is observed that the obtained profile of θ_{iia} in the intervals ①–③ are similar to the analytical plots of θ_{iia} in the intervals ①–③ shown in Fig. 5(a)–(c) respectively. The peak-peak ripple current ($\Delta I_{i,pk.,a}$) is obtained to be 3.2 A in interval ① (Fig. 26(a)), 1.7 A in interval ② (Fig. 26(b)) and 2.9 A in interval ③ (Fig. 26(c)) which are close to the calculated values of 3.39 A, 1.9 A and 3.08 A respectively (calculated using the expression of $\Delta I_{i,pk.,a}$ in Table I.).

For the 3-phase 3-wire STATCOM, $\Delta i_{i,a}$ in the intervals ① (at $\theta_{iia} = 105^\circ$), ② (at $\theta_{iia} = 135^\circ$), and ③ (at $\theta_{iia} = 165^\circ$) are shown in Fig. 27(a)–(c) respectively. As discussed in Section VI, the profile of $\Delta i_{i,a}$ in a 3-phase 3-wire STATCOM in the intervals ①, ②, and ③ are same as the profile of $\Delta i_{i,a}$ in a 3-phase 3-wire AFEC in the intervals ③, ②, and ① respectively – evident from Figs. 27(a)–(c) and 26(a)–26(c).

The experimental variation of peak-peak ripple current for a-phase ($\Delta I_{i,pk.,a}$) with θ_{iia} for the 3-phase 3-wire AFEC and the 3-phase 3-wire STATCOM is shown in Figs. 28 and 29 respectively. The experimental variation of $\Delta I_{i,pk.,a}$ with θ_{iia} is found to be close to the analytical ones for both cases.

B. $\Delta I_{i,pk.,m}$ of 3-Phase 3-Wire AFEC and 3-Phase 3-Wire STATCOM

Figs. 30 and 31 show the maximum peak-peak ripple current ($\Delta I_{i,pk.,m}$) of the 3-phase 3-wire AFEC and the 3-phase 3-wire STATCOM respectively. From Fig. 30, it is seen that $\Delta I_{i,pk.,m}$ occurs at $\theta_{iia} = 0^\circ$ for the 3-phase 3-wire AFEC as discussed in Section III. For the 3-phase 3-wire STATCOM $\Delta I_{i,pk.,m}$ occurs at $\theta_{iia} = 90^\circ$ (Fig. 31) which is discussed in Section VI. The experimental value of $\Delta I_{i,pk.,m}$ is obtained as 3.8 A for both the GCIs which is close to the expected value of 3.94 A.

C. Converter-Side Current

The hardware results in Sections XIII-A and XIII-B are obtained at $i_g = 0$ considering the assumptions (1), (2), and (3). In this section, in order to examine the variation of the ripple with $i_g \neq 0$, the converters (3-wire AFEC and 3-wire STATCOM) are operated at 75% of the rated power. The dc-bus voltage, switching frequency and grid voltages are the same as mentioned in Section XIII-A. Figs. 32 and 33 show one complete cycle of the a-phase inverter-side current (i_{ia}) along with the corresponding grid voltage (v_{ga}) and grid-side current (i_{ga}). From Figs. 32 and 28, it can be observed that the variation of peak-peak ripple

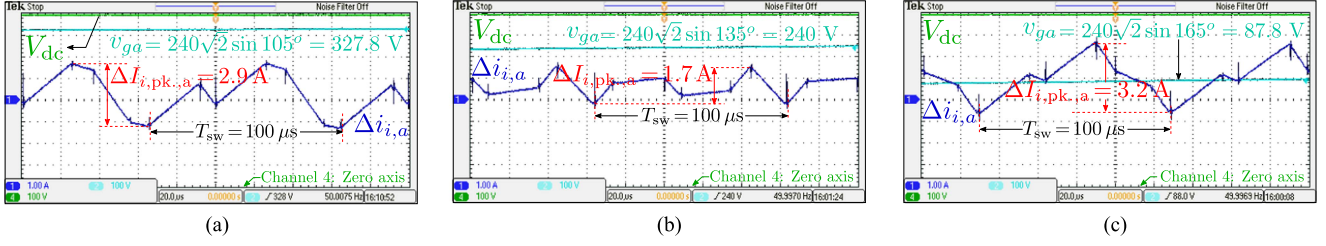


Fig. 27. Experimental result showing $\Delta i_{i,a}$ of the 3-phase 3-wire STATCOM during (a) interval ①, (b) interval ②, and (c) interval ③.

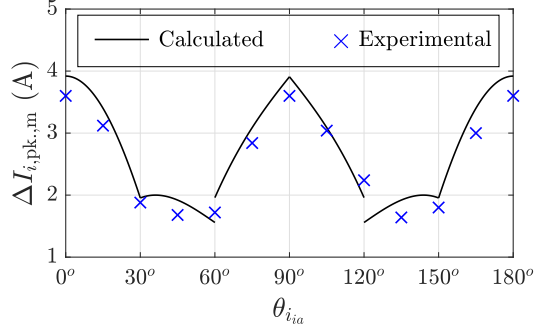


Fig. 28. Variation in $\Delta I_{i,pk.,a}$ (experimental and calculated) with $\theta_{i_{ia}}$ for the 3-phase 3-wire AFEC.

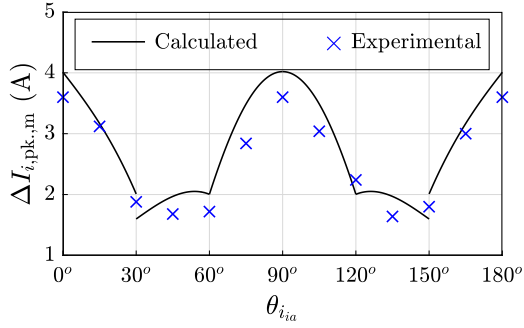


Fig. 29. Variation in $\Delta I_{i,pk.,a}$ (experimental and calculated) with $\theta_{i_{ia}}$ for the 3-phase 3-wire STATCOM.

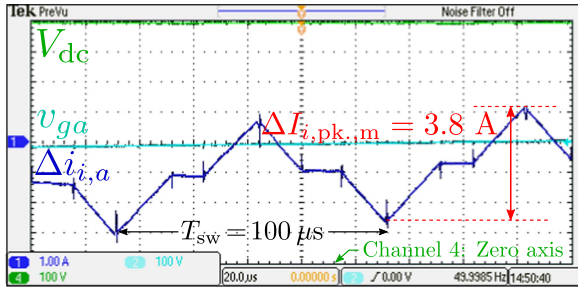


Fig. 30. Experimental result showing $\Delta I_{i,pk.,m}$ of the 3-phase 3-wire AFEC.

($\Delta I_{i,pk.}$) in i_{ia} with $\theta_{i_{ia}}$ of the 3-phase 3-wire AFEC when operating at 75% of the rated power is similar to the variation of the peak-peak ripple ($\Delta I_{i,pk.}$) in i_{ia} with $\theta_{i_{ia}}$ under zero grid-current operation. The above observation is true for the 3-phase 3-wire STATCOM too, as evident from Figs. 33 and 29.

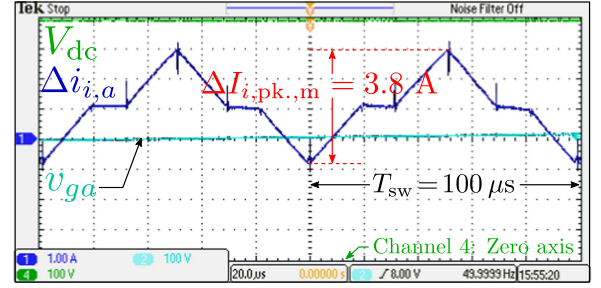


Fig. 31. Experimental result showing $\Delta I_{i,pk.,m}$ of the 3-phase 3-wire STATCOM.

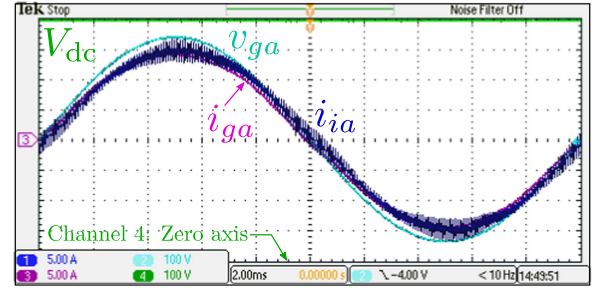


Fig. 32. Hardware result showing the inverter side current for the 3-phase 3-wire AFEC operated at 75% of the rated power.

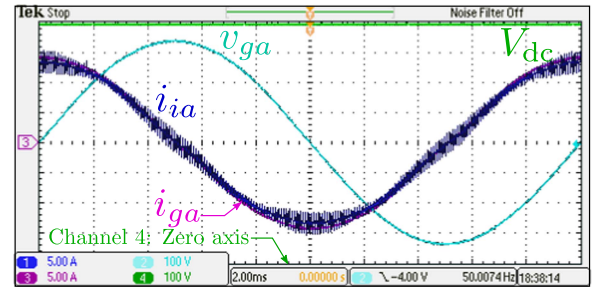


Fig. 33. Hardware result showing the inverter side current for the 3-phase 3-wire STATCOM operated at 75% of the rated power.

XIV. CONCLUSION

GCI s serve a wide range of applications like AFEC, STATCOM, and DGAs. Irrespective of the application, GCIs use an LCL filter at the output. The selection of the inverter-side inductor of the LCL filter is typically based on the maximum ripple current limit through it. In view of this, the ripple current computation is of importance. However, the ripple current varies

over every switching cycle and depends on various parameters like the dc-bus voltage, switching frequency, modulation method, application, and converter configuration. In this paper, a detailed analysis of the ripple current in the inverter side inductor in a sine-triangle modulation-based GCI for 3-wire and 4-wire configurations of AFEC, DGA, and STATCOM is presented. Closed-form expressions are developed and used to evaluate the maximum value of the peak-peak ripple current. The analysis is supported by simulation and experimental results. The analysis brings out the variation in inductor values required for different wiring configurations. The expected pattern of the ripple over a fundamental cycle of inverter current for various applications is also plotted. The developed analysis is expected to be of use in designing GCIs. After determining the value of inverter-side inductor L_i , the paper also discusses its design starting from the selection of the appropriate type of core material, the number of turns of the winding, and the required air gap length. As per the nature of the inverter-side current, an amorphous core is selected for the fabrication of L_i . An amorphous core is shown to be an apt choice for L_i due to its high saturation flux density and wide frequency range of operation. For the amorphous-core inductor design, a new area product method is proposed such that the variation in the obtained inductance is minimal with current bias. Experimental results show that the designed amorphous-core inductor meets the requirements expected of L_i in GCI applications.

REFERENCES

- [1] K. Li, A. Formentini, D. Dewar, and P. Zanchetta, "Controller design of an active front-end converter keeping in consideration grid dynamic interaction," *IEEE Trans. Ind. Electron.*, vol. 69, no. 5, pp. 5195–5206, May 2022.
- [2] D. B. Rathnayake et al., "Grid forming inverter modeling, control, and applications," *IEEE Access*, vol. 9, pp. 114781–114807, 2021.
- [3] R. H. Lasseter, Z. Chen, and D. Pattabiraman, "Grid-forming inverters: A critical asset for the power grid," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 8, no. 2, pp. 925–935, Jun. 2020.
- [4] Q. Liu, T. Caldognetto, and S. Buso, "Review and comparison of grid-tied inverter controllers in microgrids," *IEEE Trans. Power Electron.*, vol. 35, no. 7, pp. 7624–7639, Jul. 2020.
- [5] M. Liserre, F. Blaabjerg, and S. Hansen, "Design and control of an LCL-filter-based three-phase active rectifier," *IEEE Trans. Ind. Appl.*, vol. 41, no. 5, pp. 1281–1291, Sep./Oct. 2005.
- [6] *IEEE standard for interconnection and interoperability of distributed energy resources with associated electric power systems interfaces*, Standard IEEE 1547-2018 (Revision of IEEE Std 1547-2003), pp. 1–138, 2018.
- [7] J. M. Sosa, G. Escobar, P. R. Martínez-Rodríguez, G. Vázquez, M. A. Juárez, and M. Diosdado, "Comparative evaluation of L and LCL filters in transformerless grid tied converters for active power injection," in *Proc. 2014 IEEE Int. Autumn Meeting Power, Electron. Comput.*, 2014, pp. 1–6.
- [8] U. P. Yagnik and M. D. Solanki, "Comparison of L, LC & LCL filter for grid connected converter," in *Proc. 2017 Int. Conf. Trends Electron. Inform.*, 2017, pp. 455–458.
- [9] D. Solatalkaran, K. G. Khajeh, and F. Zare, "A novel filter design method for grid-tied inverters," *IEEE Trans. Power Electron.*, vol. 36, no. 5, pp. 5473–5485, May 2021.
- [10] A. Koran, S. Albatran, and J.-S. Lai, "Analytical factorized model for stability analysis and optimization of shunt RC damped LCL filter for grid-connected voltage source inverters," *IEEE Trans. Power Electron.*, vol. 35, no. 7, pp. 6830–6841, Jul. 2020.
- [11] A. Kouchaki and M. Nyman, "Analytical design of passive LCL filter for three-phase two-level power factor correction rectifiers," *IEEE Trans. Power Electron.*, vol. 33, no. 4, pp. 3012–3022, Apr. 2018.
- [12] W. Tang, K. Ma, and Y. Song, "Critical damping ratio to ensure design efficiency and stability of LCL filters," *IEEE Trans. Power Electron.*, vol. 36, no. 1, pp. 315–325, Jan. 2021.
- [13] R. N. Beres, X. Wang, M. Liserre, F. Blaabjerg, and C. L. Bak, "A review of passive power filters for three-phase grid-connected voltage-source converters," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 4, no. 1, pp. 54–69, Mar. 2016.
- [14] B. Mondal and B. A. Karuppaswamy, "Analysis of output filter inductor current ripple in grid-tie inverters," in *Proc. 2022 IEEE Int. Conf. Power Electron., Drives Energy Syst.*, 2022, pp. 1–7.
- [15] J. McLean, "Inductor design using amorphous metal C-cores," *IEEE Circuits Devices Mag.*, vol. 12, no. 5, pp. 26–30, Sep. 1996.
- [16] B. Terzić, O. Bego, M. Despalatović, G. Majić, A. Kriletić, and M. Blajić, "LCL filter design with amorphous core inductor for 100kVA energy storage converter," in *Proc. 2021 Int. Conf. Elect. Drives Power Electron.*, 2021, pp. 201–205.
- [17] R. Ren, B. Liu, Z. Dong, and F. Wang, "Current-bias dependent permeability of powder and amorphous core induced unbalanced DM impedance and mixed-mode noise," in *Proc. 2019 IEEE Energy Convers. Congr. Expo.*, 2019, pp. 2873–2880.
- [18] B. Mondal and A. Karuppaswamy B, "A noniterative design method for output LCL filter with RC damping in grid-connected inverters," *IEEE Trans. Ind. Electron.*, vol. 71, no. 12, pp. 15768–15779, Dec. 2024, doi: 10.1109/TIE.2024.3387080.
- [19] W. Xia and J. Kang, "Stability of LCL-filtered grid-connected inverters with capacitor current feedback active damping considering controller time delays," *J. Modern Power Syst. Clean Energy*, vol. 5, no. 4, pp. 584–598, Jul. 2017.
- [20] M. Saleem, K.-Y. Choi, and R.-Y. Kim, "Resonance damping for an LCL filter type grid-connected inverter with active disturbance rejection control under grid impedance uncertainty," *Int. J. Elect. Power Energy Syst.*, vol. 109, pp. 444–454, 2019.
- [21] H. Zhang, X. Wang, Y. He, D. Pan, and X. Ruan, "A compensation method to eliminate the impact of time delay on capacitor-current active damping," *IEEE Trans. Ind. Electron.*, vol. 69, no. 7, pp. 7512–7516, Jul. 2022.
- [22] Y. Tang, P. C. Loh, P. Wang, F. H. Choo, F. Gao, and F. Blaabjerg, "Generalized design of high performance shunt active power filter with output LCL filter," *IEEE Trans. Ind. Electron.*, vol. 59, no. 3, pp. 1443–1452, Mar. 2012.
- [23] M. Hanif, V. Khadkikar, W. Xiao, and J. L. Kirtley, "Two degrees of freedom active damping technique for LCL filter-based grid connected PV systems," *IEEE Trans. Ind. Electron.*, vol. 61, no. 6, pp. 2795–2803, Jun. 2014.
- [24] S. G. Parker, B. P. McGrath, and D. G. Holmes, "Regions of active damping control for LCL filters," in *Proc. 2012 IEEE Energy Convers. Congr. Expo.*, 2012, pp. 53–60.
- [25] C. W. T. McLyman, *Transformer and Inductor Design Handbook*, 4th ed. Boca Raton, FL, USA: CRC press, 2004.
- [26] W. Günther and P. Winkler, "A User Guide to Soft Magnetic Materials," in *IFe*, vol. 1, 2018, pp. 1–18.
- [27] M. Kaçki, M. S. Rylko, J. G. Hayes, and C. R. Sullivan, "Magnetic material selection for EMI filters," in *Proc. 2017 IEEE Energy Convers. Congr. Expo.*, 2017, pp. 2350–2356.
- [28] Hitachi Metals America Ltd., "Magnetization curves, powerlite inductor cores." [Online]. Available: https://www.hilltech.com/pdf/Hitachi/Datasheets/POWERLITE_C-Cores_Magnetization_Curves.pdf
- [29] B. Parikshith, "Integrated approach to filter design for grid-connected power converters," M.Sc. (Engg.) thesis, Indian Institute of Science, Bangalore, India, 2009.
- [30] M. R. Kiran, M. R. Islam, K. M. Muttaqi, and D. Sutanto, "Characterization of amorphous soft magnetic materials for toroidal core multi-winding medium frequency transformers," in *Proc. 2017 IEEE Region 10 Humanitarian Technol. Conf.*, 2017, pp. 470–473.
- [31] K. G. N. B. Abeywickrama, T. Daszczyński, Y. V. Serdyuk, and S. M. Gubanski, "Determination of complex permeability of silicon steel for use in high-frequency modeling of power transformers," *IEEE Trans. Magn.*, vol. 44, no. 4, pp. 438–444, Apr. 2008.